ECON 5280

Causal Inference and Machine Learning

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Revised lecture notes

How to Use These Notes

These notes are the canonical, print-ready version of the course. The companion Quarto Live project in the `live/` directory contains browser-executable R cells. It requires no local R installation: code runs through WebR inside the browser. Computationally intensive examples retain their full specification in the notes but use smaller classroom settings in the live pages. A native R Codespaces configuration is included as a fallback for full bootstrap inference and other heavy calculations.

The appendices preserve mathematical and computational details that are useful for reference but are not prerequisites for the later causal-inference chapters.

Contents

How to Use These Notes **i**

1 Introduction to Econometrics **1**
1.1 What is econometrics? . . . 1
1.2 An econometric description of data . . . 1
1.2.1 Variables and observations . . . 1
1.2.2 A probabilistic perspective . . . 2
1.2.3 Types of data . . . 2
1.3 Causal inference and prediction . . . 3

2 Matrix Algebra **4**
2.1 Why matrices? . . . 4
2.2 Scalars, vectors, and matrices . . . 4
2.3 Matrix multiplication . . . 4
2.3.1 Inner and outer products . . . 5
2.3.2 Rules worth remembering . . . 5
2.4 Linear independence, rank, and invertibility . . . 5
2.5 Positive semidefinite matrices . . . 6

3 Probability **7**
3.1 Random variables and distributions . . . 7
3.1.1 Joint, marginal, and conditional distributions . . . 7
3.2 Expectation and dispersion . . . 8
3.3 Conditional expectation . . . 9
3.4 Random samples and large-sample theory . . . 9
3.4.1 Convergence in probability and the WLLN . . . 9
3.4.2 Convergence in distribution and the CLT . . . 10
3.4.3 Continuous mapping and Slutsky's theorem . . . 10

4 Statistics and Regression **11**
4.1 Identification, estimation, and inference . . . 11
4.2 The linear model and least squares . . . 11
4.2.1 A method-of-moments estimator . . . 11
4.2.2 Fitted values and residuals . . . 12
4.3 Statistical properties of OLS . . . 12
4.3.1 Unbiasedness . . . 12
4.3.2 Consistency . . . 13
4.3.3 Asymptotic normality . . . 13
4.3.4 Heteroskedasticity-robust standard errors . . . 14
4.4 Testing . . . 14
4.5 Confidence intervals . . . 14
4.6 Regression in causal designs . . . 15

5 Causal Inference and Randomized Controlled Trials 16
5.1 Potential outcomes . . . 16
5.2 The average treatment effect . . . 16
5.3 Randomization and identification . . . 17
5.4 Difference in means . . . 17
5.4.1 Difference in means as OLS . . . 17
5.4.2 A survey-wording experiment . . . 18
5.5 Regression adjustment . . . 18
5.5.1 Why the interacted adjustment improves precision . . . 19
5.5.2 Monte Carlo illustration . . . 20
5.6 Summary . . . 20

6 Causal Forests 21
6.1 CATE and treatment-effect heterogeneity . . . 21
6.1.1 Unconfoundedness, overlap, and identification . . . 21
6.2 Traditional nonparametric methods . . . 22
6.2.1 Local methods . . . 22
6.2.2 Global methods . . . 22
6.3 Regression trees, causal trees, and causal forests . . . 23
6.3.1 A regression-tree sketch . . . 23
6.3.2 From a regression tree to a causal tree . . . 23
6.3.3 Honesty . . . 25
6.3.4 Causal forests . . . 25
6.3.5 Why ordinary linear regression is not enough . . . 25
6.4 Implementation with `grf` . . . 26
6.5 Summary . . . 27

7 Doubly Robust Learning 28
7.1 ATE under unconfoundedness and the failure of OLS . . . 28
7.2 The propensity score and two representations of the ATE . . . 29
7.2.1 The balancing-score result . . . 29
7.2.2 Inverse propensity weighting . . . 29
7.3 The AIPW score and double robustness . . . 30
7.4 Cross-fitting and double machine learning . . . 30
7.5 Implementation with generalized random forests . . . 31
7.6 Summary . . . 32

8 Sharp Regression Discontinuity Designs 33
8.1 A deterministic treatment rule . . . 33
8.2 Identification at the cutoff . . . 34
8.3 Local linear estimation . . . 34
8.3.1 Covariates . . . 35
8.3.2 Why not a high-order global polynomial? . . . 35
8.4 Application: electoral incumbency . . . 35
8.5 R implementation . . . 36

9 Instrumental Variables and Local Average Treatment Effects 38
9.1 Noncompliance and intent to treat . . . 38
9.2 Identification of LATE . . . 39
9.2.1 A linear moment representation . . . 40

9.2.2 Conditional LATE . . . 40
9.3 Instrumental-variable estimation . . . 40
9.3.1 Weak instruments . . . 41
9.4 Where do instruments come from? . . . 41
9.4.1 Randomized offers . . . 41
9.4.2 Natural and quasi-experiments . . . 41
9.4.3 Variables excluded from the outcome equation . . . 41
9.5 R implementation . . . 42

10 Neural Networks **44**
10.1 Neural networks as function approximators . . . 44
10.2 Conditional expectations as best predictors . . . 45
10.3 From population loss to a trained network . . . 46
10.3.1 Advantages . . . 46
10.3.2 Limitations and inference . . . 46
10.4 Application: DeepIV . . . 47
10.5 A small R illustration . . . 47

11 Difference-in-Differences and Event Studies **49**
11.1 Motivation . . . 49
11.2 Two groups and two periods . . . 50
11.2.1 No anticipation . . . 50
11.2.2 Parallel trends . . . 50
11.2.3 Identification . . . 50
11.2.4 Regression representation . . . 51
11.3 Staggered treatment adoption . . . 51
11.3.1 Valid control groups . . . 51
11.3.2 Conditional parallel trends and overlap . . . 52
11.4 Why conventional TWFE can fail . . . 52
11.4.1 Conventional event-study regressions . . . 53
11.5 Modern DID and event studies . . . 53
11.5.1 Aggregating group–time effects . . . 53
11.6 Credibility and inference . . . 54
11.6.1 A design checklist . . . 54
11.7 R implementation . . . 55
11.7.1 Generate the panel . . . 55
11.7.2 Calculate one group–time DID by hand . . . 55
11.7.3 Callaway–Sant'Anna group–time effects . . . 56
11.7.4 Aggregate the effects . . . 56
11.7.5 Conventional and interaction-weighted event studies . . . 57

A Additional Matrix Algebra **59**
A.1 Addition and scalar multiplication . . . 59
A.2 Several views of multiplication . . . 59
A.3 Rank and inverse formulas . . . 60
A.4 Eigenvalues and spectral decomposition . . . 60
A.5 Matrix derivatives . . . 61
A.6 Further reading . . . 61

B Additional Probability and Asymptotic Tools **62**
B.1 Probability spaces, events, and random variables 62
B.2 CDFs, densities, and continuous distributions 62
B.3 Conditional distributions with continuous variables 63
B.4 The bivariate normal example 63
B.5 A compact WLLN and CLT simulation 64
B.6 The delta method 64

Runtime and Reproducibility **66**

Chapter 1

Introduction to Econometrics

1.1 What is econometrics?

The term *econometrics* was introduced by Ragnar Frisch, one of the founders of the Econometric Society, the first editor of *Econometrica*, and a co-recipient of the first Nobel Memorial Prize in Economic Sciences in 1969. Frisch emphasized that econometrics is not economic statistics, economic theory, or mathematics in isolation. Its strength comes from combining all three. In modern language, we might summarize the field as

$$\text{economic models} + \text{statistical methods} + \text{economic data.}$$

This chapter draws in part on Chapter 1 of Hansen's *Econometrics* (2022).

Econometrics therefore deals with data, but so do statistics and machine learning. What is distinctive about econometrics? We will answer that question partly here and, more importantly, throughout the course.

1.2 An econometric description of data

Suppose we want to know whether obtaining a master's degree would increase future salary. One tempting comparison is:

- Alice has a master's degree and earns $12,000 per month;
- Bob entered the labor market after college and earns $8,000 per month;
- therefore, a master's degree raises salary by $4,000.

This conclusion is not reliable. Alice and Bob may differ in age, gender, major, college, ability, family background, or labor-market conditions. Two people are also far too small a sample from which to draw a general conclusion. Two natural responses are to collect more characteristics for each person (more *variables*) and to collect information from more people (a larger *sample*).

1.2.1 Variables and observations

We usually use uppercase Latin letters for variables. Let Y_i be person i's monthly salary, let X_{1i} indicate whether the person has a master's degree, and let X_{2i} be age. We collect the explanatory variables in a column vector,

$$X_i = (X_{1i}, X_{2i})'.$$

Unless stated otherwise, vectors in these notes are column vectors; a prime denotes transpose.

Suppose a survey contains n participants. The sample can be written as

$$\{(Y_i, X_i) : i = 1, \ldots, n\}.$$

It has n observations and three variables. In a data matrix, observations are rows and variables are columns. The following small example makes this concrete.

```r
set.seed(5280)                  # make the demonstration reproducible
n <- 4
Y  <- exp(rnorm(n))             # simulated monthly income
X1 <- rbinom(n, 1, 0.5)         # 1 = master's degree
X2 <- sample(22:30, n, replace = TRUE)

data_matrix <- cbind(Y, X1, X2)
data_frame  <- data.frame(income = Y, MA = X1, age = X2)
data_matrix
data_frame
```

Mathematically, the data matrix is just a matrix. We will use vectors and matrices both to represent data and to derive estimators. The notation takes some getting used to, but it turns many long calculations into short ones.

1.2.2 A probabilistic perspective

Econometrics treats the sample as random. Before we ask person i about salary, the answer is unknown to us and can be represented as a draw from a distribution. After the survey, we observe a *realization* of that random variable. For four observations, the random data matrix is

$$\begin{pmatrix} Y_1 & X_{11} & X_{21} \\ Y_2 & X_{12} & X_{22} \\ Y_3 & X_{13} & X_{23} \\ Y_4 & X_{14} & X_{24} \end{pmatrix}. \tag{1.1}$$

Before sampling, its entries are random; afterward, the realized data are fixed numbers.

A useful way to organize the logic is the following.

1. Nature determines a population distribution—the "dice of God." Researchers do not know it.
2. We specify how observations will be sampled.
3. Before seeing the outcomes, we specify an *estimator* or a *test statistic*: a rule that maps any possible sample into an answer. Because the rule is a function of random observations, the estimator is itself random before the data are observed.
4. We observe one realization of the sample.
5. We apply the rule to that realization and obtain an *estimate* or a test decision.

The distinction between an estimator and an estimate is important. The estimator is the random rule; the estimate is the number produced by the observed data. In casual discussion, people also use *sample* and *data set* interchangeably, although the former can emphasize the random object and the latter its realization.

1.2.3 Types of data

Data can be classified along several dimensions.

Observational and experimental data. Observational data record outcomes generated without researcher-controlled treatment assignment: surveys, stock returns, GDP, and prices are familiar examples. Experimental data arise when a researcher deliberately randomizes an intervention. Randomization often makes causal interpretation much easier, but many economically important interventions cannot be randomized for practical, legal, or ethical reasons. We study methods for both kinds of data.

Cross-sectional, time-series, and panel data. Cross-sectional data contain many entities observed at one time, such as a one-time survey of 1,000 graduates. Time-series data follow one entity over many periods, such as quarterly GDP. Panel data follow many entities over many periods, such as an annual household survey. Much of the course uses cross-sectional notation, but panel data become central when we study difference-in-differences and event studies in Chapter 11. In a panel, observations for the same unit over time are generally dependent; later chapters account for that dependence explicitly rather than imposing i.i.d. sampling at the unit–time level.

1.3 Causal inference and prediction

Causal inference and prediction are two major goals of data analysis. Econometrics puts particular emphasis on causal questions:

- Does air pollution reduce labor productivity?
- Does smoking during pregnancy harm infant health?
- Does graduate school raise earnings?
- Does a higher minimum wage reduce employment?

Prediction instead asks for an accurate forecast, for example tomorrow's stock price, next quarter's GDP, or which product a customer is likely to buy. A model may predict well without describing what would happen under an intervention. Conversely, a credible causal design need not be the best forecasting model.

Controlled experiments provide especially transparent causal evidence, but they are not always feasible. Econometricians have therefore developed a large toolkit for recovering causal effects from observational data. This course studies designs based on randomized experiments as well as selection on observables, instrumental variables, regression discontinuity, difference-in-differences, event studies, and modern machine-learning tools.

Chapter 2

Matrix Algebra

2.1 Why matrices?

Recall that an $n \times 1$ vector Y can collect n observations of an outcome, while an $n \times k$ matrix X collects k explanatory variables for those same observations. Most estimators in this course are formulas built from Y and X, so a compact set of matrix-algebra tools will save us a great deal of work.

This chapter contains the tools needed in the main course. Additional algebra, including eigenvalues and matrix derivatives, is collected in Appendix A.

2.2 Scalars, vectors, and matrices

A *scalar* is a number. A *vector* is an ordered list of numbers, usually written as a column. A *matrix* is a rectangular array of numbers. If A has n rows and k columns, then it is an $n \times k$ matrix. Its entry in row i and column j is a_{ij} (or A_{ij}). For example,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = (A_1, A_2, A_3)$$

is 2×3, where A_j is its jth column. A scalar is a 1×1 matrix, and a column vector is a matrix with one column.

Definition 2.1 (Transpose)**.** The transpose A' is obtained by interchanging rows and columns: $(A')_{ij} = A_{ji}$. If A is $n \times k$, then A' is $k \times n$.

Thus, for the matrix above,

$$A' = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{pmatrix}.$$

Definition 2.2 (Square, symmetric, diagonal, and identity matrices)**.** A matrix is *square* if it has the same number of rows and columns. A square matrix is *symmetric* if $A = A'$. A matrix is *diagonal* if all off-diagonal entries are zero. The $k \times k$ identity matrix I_k is diagonal with every diagonal entry equal to one.

Symmetric matrices appear repeatedly as covariance matrices and quadratic forms. Identity matrices play the same role for matrix multiplication that the scalar 1 plays for ordinary multiplication.

2.3 Matrix multiplication

Two matrices A and B are *conformable* for the product AB when the number of columns of A equals the number of rows of B. If A is $m \times k$ and B is $k \times r$, then AB is $m \times r$, with

$$(AB)_{ij} = \sum_{\ell=1}^{k} a_{i\ell} b_{\ell j}. \tag{2.1}$$

Always check dimensions before multiplying.

2.3.1 Inner and outer products

For two $k \times 1$ vectors a and b, the *inner product* is the scalar

$$a'b = \sum_{j=1}^{k} a_j b_j.$$

The *outer product* ab' is instead a $k \times k$ matrix whose (i, j) entry is $a_i b_j$. This distinction matters throughout econometrics. For instance, $X_i X_i'$ is a $k \times k$ outer product, while $X_i'\beta$ is a scalar inner product.

If X is $n \times k$ and its ith row is X_i', then

$$X'X = \sum_{i=1}^{n} X_i X_i'. \tag{2.2}$$

This identity turns a sample average of outer products into compact matrix notation.

2.3.2 Rules worth remembering

Matrix multiplication is generally not commutative: even when both products exist, AB need not equal BA. It is associative and distributive whenever the dimensions conform:

$$A(BC) = (AB)C, \qquad A(B + C) = AB + AC.$$

Transpose reverses the order of a product:

$$(AB)' = B'A'.$$

```r
A <- matrix(c(1, 2, 3, 4, 5, 6), nrow = 3, ncol = 2)
B <- matrix(c(2, 3, 4, 5, 6, 7), nrow = 3, ncol = 2)
t(A) %*% B

# The same (1,2) entry as an inner product of two columns
sum(A[, 1] * B[, 2])
```

2.4 Linear independence, rank, and invertibility

Vectors $a_1, \ldots, a_r$ are *linearly independent* if

$$c_1 a_1 + \cdots + c_r a_r = 0$$

implies $c_1 = \cdots = c_r = 0$. Otherwise at least one direction is redundant.

Definition 2.3 (Rank and full rank)**.** The rank of a matrix is the dimension of its column space (equivalently, its row space). An $m \times k$ matrix has

$$\text{rank}(A) \leq \min\{m, k\}.$$

It is *full column rank* if its rank is k, *full row rank* if its rank is m, and simply *full rank* if its rank is $\min\{m, k\}$.

An inverse is defined only for a square matrix. A $k \times k$ matrix A is invertible if there exists A^{-1} satisfying

$$A^{-1}A = AA^{-1} = I_k.$$

A square matrix is invertible if and only if it has rank k. A rectangular full-rank matrix is *not* invertible, although one of its Gram matrices may be:

- $A'A$ is invertible if and only if A has full column rank;
- AA' is invertible if and only if A has full row rank.

Both Gram matrices can be invertible simultaneously only when A is square and full rank. Also,

$$\operatorname{rank}(A'A) = \operatorname{rank}(AA') = \operatorname{rank}(A).$$

If conformable square matrices A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$.

2.5 Positive semidefinite matrices

Definition 2.4 (Positive definiteness)**.** A real symmetric $k \times k$ matrix A is *positive semidefinite* (p.s.d.) if $x'Ax \geq 0$ for every $x \in \mathbb{R}^k$. It is *positive definite* (p.d.) if $x'Ax > 0$ for every nonzero x.

For any real $m \times k$ matrix A,

$$x'A'Ax = (Ax)'(Ax) = \|Ax\|^2 \geq 0,$$

so $A'A$ is p.s.d. It is p.d. precisely when A has full column rank. Similarly, AA' is p.s.d. and is p.d. precisely when A has full row rank. This argument is both shorter and more general than appealing to an eigendecomposition.

Covariance matrices are p.s.d. for the same reason. If W is a random vector with finite second moments and covariance matrix Σ, then for any a,

$$a'\Sigma a = \mathbb{E}\left[\{a'(W - \mathbb{E}W)\}^2\right] \geq 0.$$

The optional algebra in Appendix A gives detailed formulas, eigenvalue characterizations for symmetric matrices, and the matrix derivatives used to derive least squares.

Chapter 3

Probability

Econometric estimators are functions of random samples. Probability gives us the language for describing their behavior before the sample is observed. This chapter reviews the ideas used repeatedly later in the course. Optional measure-theoretic details, longer simulations, and the delta method appear in Appendix B.

3.1 Random variables and distributions

A random variable is a numerical function of an underlying random outcome. For example, the outcome might describe all features of tomorrow's weather, while X records only whether it rains. The set of values that X can take with positive probability, or can approach arbitrarily closely, is its *support*. The support should not be confused with the underlying sample space of outcomes.

A random variable is *discrete* when its distribution is concentrated on a finite or countable set. Its probability mass function (p.m.f.) is

$$p_X(x) = \Pr(X = x).$$

A Bernoulli random variable $X \sim \text{Bernoulli}(p)$ has support $\{0, 1\}$ and $\Pr(X = 1) = p$.

```r
set.seed(5280)
rbinom(n = 10, size = 1, prob = 0.5)
```

The cumulative distribution function (c.d.f.) is defined for every random variable:

$$F_X(x) = \Pr(X \le x), \qquad x \in \mathbb{R}.$$

If X has a density f_X, then

$$F_X(x) = \int_{-\infty}^{x} f_X(s)\,ds, \qquad \Pr(a < X \le b) = \int_a^b f_X(s)\,ds.$$

Such a variable is called *absolutely continuous*, and $\Pr(X = x) = 0$ for every x. Probability is not generally the geometric length of an interval: it is the area under the density. Length gives probability only for a suitably normalized uniform distribution.

A normal variable $X \sim N(\mu, \sigma^2)$ has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.$$

Normal distributions are important for large-sample approximations; we will not assume that economic variables themselves are normally distributed.

3.1.1 Joint, marginal, and conditional distributions

The joint distribution of (X, Y) describes their behavior together. Its marginal distributions describe X and Y separately. Conditional distributions describe one variable within a subpopulation indexed by the other.

When a joint density exists,

$$f_X(x) = \int f_{X,Y}(x,y)\,dy, \qquad f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

for values of x with $f_X(x) > 0$. Hence

$$f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x) = f_{X|Y}(x \mid y) f_Y(y).$$

The joint distribution contains the marginals, but the two marginals alone usually do not reveal how X and Y move together.

Definition 3.1 (Independence)**.** X and Y are independent, written $X \perp Y$, if

$$F_{X,Y}(x,y) = F_X(x)F_Y(y)$$

for every (x,y). When densities exist, this is equivalent to $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ almost everywhere.

If $X \perp Y$, then $g(X) \perp h(Y)$ for measurable functions g and h.

3.2 Expectation and dispersion

For a discrete variable, replace the integrals below by sums. Provided the relevant integral exists, the expectation is

$$\mathbb{E}(X) = \int x f_X(x)\,dx = \mu_X.$$

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y),$$

whether or not X and Y are independent.

Variance and covariance are

$$\begin{aligned} \operatorname{Var}(X) &= \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}(X^2) - \mu_X^2, \\ \operatorname{Cov}(X,Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}(XY) - \mu_X\mu_Y. \end{aligned}$$

Useful rules include

$$\begin{aligned} \operatorname{Var}(aX + bY) &= a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X,Y), \\ \operatorname{Cov}(aX + bY, Z) &= a \operatorname{Cov}(X,Z) + b \operatorname{Cov}(Y,Z). \end{aligned}$$

The correlation is

$$\rho_{XY} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} \in [-1,1]$$

when both variances are positive. Correlation is unit-free, but zero correlation need not imply independence.

For a $k \times 1$ random vector X, its mean is the vector $\mu_X = \mathbb{E}(X)$ and its covariance matrix is

$$\Sigma_X = \operatorname{Var}(X) = \mathbb{E}[(X - \mu_X)(X - \mu_X)'].$$

It is symmetric and p.s.d. because, for every $a \in \mathbb{R}^k$,

$$a'\Sigma_X a = \operatorname{Var}(a'X) \geq 0.$$

3.3 Conditional expectation

Conditional expectation is among the most important objects in this course. For a continuous pair with a conditional density,

$$\mathbb{E}(Y \mid X = x) = \int y f_{Y|X}(y \mid x)\, dy.$$

For a fixed x, this is a number. By contrast, $\mathbb{E}(Y \mid X)$ is a random variable—a function of the still-random X.

Conditional expectation is linear, and variables measurable with respect to the conditioning information can be taken outside:

$$\begin{aligned}\mathbb{E}(aY + bZ \mid X) &= a\mathbb{E}(Y \mid X) + b\mathbb{E}(Z \mid X),\\ \mathbb{E}[g(X)Y \mid X] &= g(X)\mathbb{E}(Y \mid X).\end{aligned}$$

Theorem 3.1 (Law of iterated expectations)**.** *If the expectations exist, then*

$$\mathbb{E}\{\mathbb{E}(Y \mid X)\} = \mathbb{E}(Y).$$

More generally, if the information in Z is contained in the information in X, then

$$\mathbb{E}\{\mathbb{E}(Y \mid X) \mid Z\} = \mathbb{E}(Y \mid Z).$$

In particular, $\mathbb{E}(X \mid X) = X$. We say that Y is *mean independent* of X if $\mathbb{E}(Y \mid X) = \mathbb{E}(Y)$ almost surely. Mean independence implies zero covariance (when second moments exist), but it is not symmetric in general. Full independence implies mean independence in both directions:

$$X \perp Y \implies \mathbb{E}(Y \mid X) = \mathbb{E}(Y) \implies \mathrm{Cov}(X, Y) = 0.$$

3.4 Random samples and large-sample theory

For cross-sectional applications, we often assume that $W_i = (Y_i, X_i')'$ is independent and identically distributed across units i. This means $W_i \perp W_j$ for $i \neq j$ and each unit has the same distribution. It does *not* require the components within W_i to be independent. In panel data, repeated observations on the same unit are normally dependent; i.i.d. sampling may instead be imposed across units or replaced by a suitable dependence condition.

3.4.1 Convergence in probability and the WLLN

Definition 3.2 (Convergence in probability)**.** A sequence of random vectors W_n converges in probability to w, written $W_n \xrightarrow{p} w$, if, for every $\varepsilon > 0$,

$$\Pr(\|W_n - w\| > \varepsilon) \longrightarrow 0.$$

When W_n is an estimator and w is the target parameter, this property is called consistency.

Theorem 3.2 (Weak law of large numbers)**.** *Let $X_1, \ldots, X_n$ be i.i.d. random vectors with $\mathbb{E}\|X_i\| < \infty$. Then*

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mathbb{E}(X_i).$$

For a scalar random variable with finite variance, the intuition is especially transparent because

$$\mathrm{Var}(\bar{X}_n) = \frac{\mathrm{Var}(X_i)}{n} \longrightarrow 0.$$

3.4.2 Convergence in distribution and the CLT

Definition 3.3 (Convergence in distribution)**.** W_n converges in distribution to W, written $W_n \xrightarrow{d} W$, if

$$\mathbb{E}[g(W_n)] \longrightarrow \mathbb{E}[g(W)]$$

for every bounded continuous function g. For scalar variables this is equivalent to $F_{W_n}(w) \to F_W(w)$ at every continuity point of F_W.

It is not correct to require convergence of probabilities for *every* set: boundary sets can fail even when convergence in distribution holds.

Theorem 3.3 (Multivariate central limit theorem)**.** *Let $X_1, \ldots, X_n$ be i.i.d. $k \times 1$ random vectors with mean μ and finite covariance matrix Σ. Then*

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma).$$

The limiting distribution is standard normal only after appropriate standardization. In the multivariate result, Σ contains all cross-component covariances, not merely the marginal variances.

3.4.3 Continuous mapping and Slutsky's theorem

If g is continuous at the relevant limiting values, the continuous mapping theorem gives

$$W_n \xrightarrow{p} W \Longrightarrow g(W_n) \xrightarrow{p} g(W),$$

and an analogous statement holds for convergence in distribution.

Slutsky's theorem says that if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ for a constant c, then

$$\begin{aligned}
X_n + Y_n &\xrightarrow{d} X + c, \\
X_n Y_n &\xrightarrow{d} cX, \\
X_n / Y_n &\xrightarrow{d} X/c \quad (c \neq 0).
\end{aligned}$$

These tools turn probability limits and central limit theorems into the large-sample distributions of econometric estimators.

Chapter 4

Statistics and Regression

Many causal estimators studied later can be represented as linear regressions or as solutions to moment equations. This chapter develops least squares, its large-sample properties, heteroskedasticity-robust standard errors, tests, and confidence intervals.

4.1 Identification, estimation, and inference

From a frequentist perspective, data are generated by a fixed but unknown process. For example,

$$Y = g(X, \varepsilon),$$

together with the distribution of (X, ε), describes a data generating process (DGP). We observe a sample of (Y_i, X_i) but not the structural disturbance ε_i or the function g.

Suppose the target is a feature θ of the DGP. Econometric analysis has three distinct tasks:

Identification. Does the population distribution determine a unique value of θ under the maintained assumptions?

Estimation. Can we construct a sample rule $\widehat{\theta}$ that is close to θ?

Inference. How much sampling uncertainty surrounds $\widehat{\theta}$? Tests and confidence intervals answer related but different versions of this question.

No amount of data repairs a failure of identification: if two parameter values produce the same observable distribution, the sample cannot tell them apart.

4.2 The linear model and least squares

Consider an i.i.d. sample satisfying the population linear projection

$$Y_i = X_i'\beta + \varepsilon_i, \qquad \mathbb{E}(X_i\varepsilon_i) = 0, \tag{4.1}$$

where X_i is $k \times 1$ and usually contains a constant. Stacking observations gives

$$Y = X\beta + \varepsilon,$$

where X is $n \times k$ and row i is X_i'. At this stage, β need not have a causal interpretation; later designs supply the assumptions that make particular regression coefficients causal.

4.2.1 A method-of-moments estimator

The orthogonality condition in (4.1) implies

$$\mathbb{E}(X_iY_i) = \mathbb{E}(X_iX_i')\beta.$$

If $Q = \mathbb{E}(X_iX_i')$ is nonsingular, then the population distribution identifies

$$\beta = Q^{-1}\mathbb{E}(X_iY_i). \tag{4.2}$$

Nonsingularity rules out perfect multicollinearity. For example, a constant, a male indicator, and a female indicator cannot all be included when the two indicators always sum to one. One of the three columns must be removed.

The sample analogue replaces expectations by sample averages:

$$\begin{aligned}\widehat{\beta} &= \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} X_i Y_i\right) \\ &= (X'X)^{-1}X'Y. \end{aligned} \tag{4.3}$$

This is both a method-of-moments estimator and the ordinary least-squares (OLS) estimator. The least-squares interpretation follows because (4.3) minimizes $\sum_i (Y_i - X_i'b)^2$ when X has full column rank.

Method of moments is a broad and useful recipe:

1. obtain population moment restrictions;
2. replace population moments by sample averages;
3. solve the sample moment equations.

It does not eliminate the hard work: credible moment restrictions still come from economic reasoning and an identification argument.

```r
set.seed(5280)
n <- 100
x <- rnorm(n)
e <- rnorm(n)
y <- 0.5 + 2 * x + e
X <- cbind(1, x)

# Matrix formula
bhat <- solve(crossprod(X), crossprod(X, y))
bhat

# R's regression routine
coef(lm(y ~ x))
```

4.2.2 Fitted values and residuals

The fitted values and residuals are

$$\begin{aligned}\widehat{Y} &= X\widehat{\beta} = X(X'X)^{-1}X'Y \equiv PY, \\ \widehat{\varepsilon} &= Y - \widehat{Y} = \{I - X(X'X)^{-1}X'\}Y \equiv MY.\end{aligned}$$

The matrices P and $M = I - P$ project Y respectively onto the column space of X and its orthogonal complement. The normal equations imply $X'\widehat{\varepsilon} = 0$.

4.3 Statistical properties of OLS

4.3.1 Unbiasedness

Definition 4.1 (Unbiasedness)**.** An estimator $\widehat{\beta}$ is unbiased for β if $\mathbb{E}(\widehat{\beta}) = \beta$.

Unbiasedness is a finite-sample property. For conditional unbiasedness of OLS, assume

$$\mathbb{E}(\varepsilon \mid X) = 0. \tag{4.4}$$

Under random sampling, the familiar condition $\mathbb{E}(\varepsilon_i \mid X_i) = 0$ implies (4.4). Conditional on a full-rank design matrix,

$$\begin{aligned}\mathbb{E}(\widehat{\beta} \mid X) &= \mathbb{E}\{\beta + (X'X)^{-1}X'\varepsilon \mid X\} \\ &= \beta + (X'X)^{-1}X'\mathbb{E}(\varepsilon \mid X) \\ &= \beta.\end{aligned}$$

The law of iterated expectations therefore gives $\mathbb{E}(\widehat{\beta}) = \beta$. Notice that the target is β, not zero.

```r
set.seed(5280)
n <- 100
B <- 500
bhat <- replicate(B, {
  x <- rnorm(n)
  y <- 0.5 + 2 * x + rnorm(n)
  coef(lm(y ~ x))
})
rowMeans(bhat)
```

4.3.2 Consistency

Using $Y = X\beta + \varepsilon$,

$$\widehat{\beta} - \beta = \left(\frac{1}{n}\sum_i X_i X_i'\right)^{-1} \left(\frac{1}{n}\sum_i X_i \varepsilon_i\right). \tag{4.5}$$

Suppose the relevant moments exist, observations are i.i.d., $Q = \mathbb{E}(X_i X_i')$ is nonsingular, and $\mathbb{E}(X_i \varepsilon_i) = 0$. The WLLN and continuous mapping theorem give

$$\frac{1}{n}\sum_i X_i X_i' \xrightarrow{p} Q, \qquad \frac{1}{n}\sum_i X_i \varepsilon_i \xrightarrow{p} 0,$$

and hence $\widehat{\beta} \xrightarrow{p} \beta$. Conditional mean zero is sufficient but not necessary for this conclusion; the weaker orthogonality condition is enough.

4.3.3 Asymptotic normality

Multiply (4.5) by $\sqrt{n}$:

$$\sqrt{n}(\widehat{\beta} - \beta) = \left(\frac{1}{n}\sum_i X_i X_i'\right)^{-1} \left(\frac{1}{\sqrt{n}}\sum_i X_i \varepsilon_i\right).$$

Let

$$Q = \mathbb{E}(X_i X_i'), \qquad \Omega = \mathbb{E}(\varepsilon_i^2 X_i X_i').$$

Under the preceding conditions and finite second moments for $X_i \varepsilon_i$, the multivariate CLT and Slutsky's theorem yield

$$\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, \Sigma), \qquad \Sigma = Q^{-1}\Omega Q^{-1}. \tag{4.6}$$

The matrix $Q^{-1}\Omega Q^{-1}$ is called a sandwich variance: Q^{-1} is the bread and Ω is the filling. Equivalently, for large n, $\widehat{\beta}$ is approximately $N(\beta, \Sigma/n)$.

4.3.4 Heteroskedasticity-robust standard errors

Neither consistency nor (4.6) requires constant conditional error variance. A heteroskedasticity-robust estimator of Σ is

$$\widehat{\Sigma} = \widehat{Q}^{-1}\widehat{\Omega}\widehat{Q}^{-1}, \quad \widehat{Q} = \frac{1}{n}\sum_i X_i X_i', \quad \widehat{\Omega} = \frac{1}{n}\sum_i \widehat{\varepsilon}_i^2 X_i X_i'. \tag{4.7}$$

If $\widehat{\Sigma}_{jj}$ is diagonal element j, the standard error of $\widehat{\beta}_j$ is

$$\mathrm{se}(\widehat{\beta}_j) = \sqrt{\widehat{\Sigma}_{jj}/n}.$$

It estimates the standard deviation of $\widehat{\beta}_j$, not the standard deviation of the limiting variable in (4.6).

4.4 Testing

Consider

$$H_0 : \beta_j = \beta_j^0 \qquad \text{against} \qquad H_1 : \beta_j \neq \beta_j^0.$$

A Type I error rejects a true null; a Type II error fails to reject a false null. A level-α procedure controls the probability of a Type I error at approximately α. Its power is the probability of rejection when the alternative is true.

Under H_0, the studentized statistic

$$T_n = \frac{\widehat{\beta}_j - \beta_j^0}{\mathrm{se}(\widehat{\beta}_j)} \xrightarrow{d} N(0, 1). \tag{4.8}$$

For a two-sided asymptotic level-α test, reject when $|T_n| > z_{1-\alpha/2}$. Common critical values are 1.645 for 10 percent, 1.960 for 5 percent, and 2.576 for 1 percent tests.

The two-sided p-value is

$$2\{1 - \Phi(|t_{\mathrm{obs}}|)\}.$$

It is the null probability of observing a statistic at least as extreme as the realized one. It is *not* the probability that the null hypothesis is true. Failure to reject should not be described as acceptance: it may reflect low power.

Because standard errors shrink with sample size, a very small effect can be statistically significant in a very large sample. Statistical significance must therefore be distinguished from economic importance.

4.5 Confidence intervals

An asymptotic $100(1-\alpha)$ percent confidence interval is

$$\left[\widehat{\beta}_j - z_{1-\alpha/2}\,\mathrm{se}(\widehat{\beta}_j), \widehat{\beta}_j + z_{1-\alpha/2}\,\mathrm{se}(\widehat{\beta}_j)\right]. \tag{4.9}$$

Its coverage probability approaches $1-\alpha$ under the maintained assumptions. The parameter is fixed; across repeated samples, the random interval changes. A 95 percent procedure covers the fixed true parameter in approximately 95 percent of repeated samples. It does not assign a 95 percent probability to the parameter after a particular frequentist interval has been observed.

The interval and the two-sided test are dual: a candidate value is rejected at level α exactly when it lies outside the corresponding confidence interval (up to numerical and approximation differences). Reporting point estimates with standard errors or confidence intervals conveys both magnitude and uncertainty.

```r
library(sandwich)
library(lmtest)

set.seed(5280)
n <- 500
x1 <- rnorm(n)
x2 <- rnorm(n)
y <- 0.5 + x1 + x2 + rnorm(n, sd = 0.5 + abs(x1))
fit <- lm(y ~ x1 + x2)

# HC1 heteroskedasticity-robust covariance matrix and coefficient table
V_HC1 <- vcovHC(fit, type = "HC1")
coeftest(fit, vcov. = V_HC1)

# A 95% robust confidence interval for the coefficient on x1
b1 <- coef(fit)["x1"]
se1 <- sqrt(V_HC1["x1", "x1"])
b1 + c(-1, 1) * qnorm(0.975) * se1
```

4.6 Regression in causal designs

We will almost never assume without argument that a regression coefficient is a causal effect. Instead, experiments or observational research designs will produce identifying comparisons, and regression will often provide a convenient way to compute them. Always separate the statistical question “How is this coefficient estimated?” from the causal question “Why does this coefficient equal the effect of interest?”

Chapter 5

Causal Inference and Randomized Controlled Trials

Outline

This chapter introduces the potential-outcomes language of causal inference, the average treatment effect, and randomized controlled trials (RCTs). We then study two estimators: the difference in means and regression adjustment. The second method uses baseline covariates to improve precision without requiring the outcome to be linear in those covariates.

5.1 Potential outcomes

There is more than one way to formalize causality. We use the potential-outcomes framework, associated with Neyman and Rubin. Imagine an individual indexed by i.

- The *treatment* D_i records the treatment that individual i receives. Treatments may be discrete or continuous, but here $D_i \in \{0, 1\}$.
- The *potential outcome* $Y_i(d)$ is the outcome that would be observed for individual i under treatment status d. For example, $Y_i(1)$ could be future income if i attends HKUST and $Y_i(0)$ future income otherwise.
- The observed outcome satisfies the *consistency relation*

$$Y_i = Y_i(D_i) = D_i Y_i(1) + (1 - D_i) Y_i(0). \tag{5.1}$$

Writing $Y_i(d)$ presumes that the treatment is well defined and that one individual's potential outcome is unaffected by other individuals' treatment assignments. Together with consistency, these conditions are commonly grouped under the *stable unit treatment value assumption* (SUTVA). SUTVA rules out, for example, spillovers between treated and untreated students. Whether it is credible depends on the application.

The individual treatment effect is

$$Y_i(1) - Y_i(0).$$

It is mathematically well defined, but both potential outcomes cannot normally be observed for the same individual. This is the *fundamental problem of causal inference*. Replacing the missing $Y_i(0)$ by another person's observed outcome would require those two people to have the same untreated potential outcome—an implausibly strong requirement.

5.2 The average treatment effect

We therefore begin with an average causal parameter:

$$\tau \equiv \text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]. \tag{5.2}$$

Under i.i.d. sampling, this population expectation does not depend on the label i. Notice that treatment effects may still differ across individuals; the ATE summarizes their population average.

5.3 Randomization and identification

In an RCT, the researcher assigns treatment at random. In the simple Bernoulli design used for the population calculations below,

$$D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0)\big), \qquad \Pr(D_i = 1) = \pi \in (0,1). \tag{5.3}$$

For example, subjects in a drug trial may be randomly assigned either the drug or a placebo. Assignment then does not depend on baseline health, income, or the outcomes that the subjects would experience under either arm. A survey that merely compares people who chose to take the drug with people who did not would not, in general, have this property.

Randomization turns the unobserved potential-outcome means into observable conditional means:

$$\begin{aligned}
\tau &= \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] \\
&= \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \\
&= \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0].
\end{aligned} \tag{5.4}$$

The second line uses random assignment, and the third uses consistency. Equation (5.4) *identifies* the ATE: it expresses a causal parameter in terms of the distribution of observed variables.

5.4 Difference in means

Let $n_d = \sum_{i=1}^n \mathbf{1}\{D_i = d\}$. The sample analogue of (5.4) is

$$\widehat{\tau}_{\mathrm{DM}} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i. \tag{5.5}$$

Conditional on having observations in both arms, this estimator is unbiased under random assignment. It is also consistent and asymptotically normal. If treatment is assigned independently with probability π, then

$$\sqrt{n}\big(\widehat{\tau}_{\mathrm{DM}} - \tau\big) \xrightarrow{d} \mathcal{N}(0, V_{\mathrm{DM}}), \qquad V_{\mathrm{DM}} = \frac{\mathrm{Var}(Y_i(1))}{\pi} + \frac{\mathrm{Var}(Y_i(0))}{1-\pi}. \tag{5.6}$$

Equivalently, the two variances in (5.6) can be written as $\mathrm{Var}(Y_i \mid D_i = d)$. A heteroskedasticity-robust standard error estimates the sampling uncertainty without imposing equal outcome variances across the two arms.

5.4.1 Difference in means as OLS

Because D is binary,

$$\begin{aligned}
\mathbb{E}[Y \mid D] &= \mathbb{E}[Y \mid D = 0] + D\{\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]\} \\
&= \delta_0 + \tau D,
\end{aligned}$$

where the last equality uses (5.4). Defining $\varepsilon = Y - \mathbb{E}[Y \mid D]$ gives

$$Y_i = \delta_0 + \tau D_i + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid D_i] = 0. \tag{5.7}$$

This is a population projection identity, not an assumption that the outcome has a structurally linear model.

The OLS slope from (5.7) is exactly

$$\begin{aligned}\widehat{\tau} &= \frac{\sum_i (D_i - \bar{D}) Y_i}{\sum_i (D_i - \bar{D})^2} \\ &= \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i = \widehat{\tau}_{\mathrm{DM}}.\end{aligned}$$

Thus a regression of Y on an intercept and D is simply a convenient way to compute the difference in means and its robust standard error.

5.4.2 A survey-wording experiment

The following example, adapted from the machine-learning causal-inference tutorial by Susan Athey and Stefan Wager, uses a simplified General Social Survey experiment. Roughly half the respondents were asked whether the government spends too much on "welfare" ($D_i = 1$); the others were asked about "assistance to the poor" ($D_i = 0$). The binary outcome equals one for a "yes" response. The code verifies that the direct calculation and OLS produce the same point estimate.

```r
library(lmtest)
library(sandwich)

gss <- read.csv("data/welfare-small.csv")
Y <- gss$Y
D <- gss$D

# Direct difference in means
tau_dm <- mean(Y[D == 1]) - mean(Y[D == 0])
tau_dm

# The coefficient on D is exactly the same number
fit_dm <- lm(Y ~ D)
coeftest(fit_dm, vcov. = vcovHC(fit_dm, type = "HC2"))
```

The wording produces a statistically detectable difference in respondents' interpretation of government spending.

5.5 Regression adjustment

The difference in means ignores baseline covariates that predict outcomes. Let W_i contain such pre-treatment variables and strengthen (5.3) to

$$D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0), W_i\big). \tag{5.8}$$

Randomization makes covariate adjustment unnecessary for identification, but useful for precision.

The safest general-purpose specification allows the relationship between Y and W to differ by treatment arm. Center W_i at its population mean, $X_i = W_i - \mathbb{E}[W_i]$, and define the arm-specific linear-projection slopes

$$\gamma_d = \{\mathbb{E}[X_i X_i']\}^{-1}\mathbb{E}[X_i\{Y_i(d) - \mathbb{E}[Y_i(d)]\}], \qquad d \in \{0,1\}, \tag{5.9}$$

assuming $\mathbb{E}[X_i X_i']$ is nonsingular. These slopes always exist under the stated moment conditions. They do *not* assert that the conditional mean of $Y_i(d)$ is linear.

In practice, replace $\mathbb{E}[W_i]$ by the overall sample mean $\bar{W}$ and estimate the fully interacted regression

$$Y_i = \alpha + \tau D_i + (W_i - \bar{W})'\beta_0 + D_i(W_i - \bar{W})'(\beta_1 - \beta_0) + u_i. \tag{5.10}$$

The coefficient on D_i compares the fitted treatment and control outcomes at the overall covariate mean. It consistently estimates the ATE under randomization, even if the fitted linear regressions are misspecified. Use a heteroskedasticity-robust standard error. The estimator need not be exactly unbiased in a finite sample, but its bias is asymptotically negligible.

5.5.1 Why the interacted adjustment improves precision

This argument is worth doing carefully. Define the arm-specific projection residuals

$$r_i(d) = Y_i(d) - \mathbb{E}[Y_i(d)] - X_i'\gamma_d.$$

The normal equations imply $\mathbb{E}[X_i r_i(d)] = 0$, so

$$\mathrm{Var}(Y_i(d)) = \mathrm{Var}(r_i(d)) + \gamma_d'\mathbb{E}[X_i X_i']\gamma_d. \tag{5.11}$$

Because the regression is evaluated at the overall sample covariate mean, sampling variation in that mean also enters when the two arm-specific slopes differ. The interacted regression-adjustment estimator has asymptotic variance

$$V_{\mathrm{RA}} = \frac{\mathrm{Var}(r_i(1))}{\pi} + \frac{\mathrm{Var}(r_i(0))}{1-\pi} + (\gamma_1 - \gamma_0)'\mathbb{E}[X_i X_i'](\gamma_1 - \gamma_0). \tag{5.12}$$

Combining (5.6), (5.11), and (5.12) yields

$$V_{\mathrm{DM}} - V_{\mathrm{RA}} = \left\{\sqrt{\frac{1-\pi}{\pi}}\gamma_1 + \sqrt{\frac{\pi}{1-\pi}}\gamma_0\right\}'\mathbb{E}[X_i X_i']\left\{\sqrt{\frac{1-\pi}{\pi}}\gamma_1 + \sqrt{\frac{\pi}{1-\pi}}\gamma_0\right\} \geq 0. \tag{5.13}$$

The gain is large when baseline covariates explain substantial outcome variation. No invalid equality between treatment-arm conditional covariances is needed: each arm has its own projection slope and residual variance.

A regression without treatment–covariate interactions is still consistent for the ATE under (5.8). With unequal treatment probabilities and heterogeneous covariate slopes, however, it does not have the same general no-loss-of-precision guarantee. The centered, fully interacted specification in (5.10) is therefore the preferred default.

For the survey experiment, the implementation is:

```
covariates <- c("age", "polviews", "income", "educ", "marital", "sex")
gss_c <- gss
gss_c[covariates] <- lapply(gss[covariates], function(x) x - mean(x))

fit_ra <- lm(
  Y ~ D * (age + polviews + income + educ + marital + sex),
  data = gss_c
)
coeftest(fit_ra, vcov. = vcovHC(fit_ra, type = "HC2"))["D", ]
```

Only pre-treatment covariates should be used. Adjusting for variables caused by treatment can change the estimand and introduce bias.

5.5.2 Monte Carlo illustration

The next experiment uses a deliberately nonlinear outcome model. Because D is randomized independently of W, the true ATE is

$$\mathbb{E}[1 + W^2] = 1 + \{\mathrm{Var}(W) + \mathbb{E}[W]^2\} = 6$$

when $W \sim \mathcal{N}(2, 1)$. Both estimators are consistent; the interacted adjustment uses the predictive information in W and is more precise.

```r
set.seed(5280)
n <- 1000
B <- 500
tau_dm <- numeric(B)
tau_ra <- numeric(B)

for (b in seq_len(B)) {
  W <- rnorm(n, 2, 1)
  D <- rbinom(n, 1, 0.5)
  e <- rnorm(n)
  Y <- 1 + D + W^2 * D + exp(W) + e

  tau_dm[b] <- coef(lm(Y ~ D))["D"]
  Wc <- W - mean(W)
  tau_ra[b] <- coef(lm(Y ~ D * Wc))["D"]
}

c(mean_dm = mean(tau_dm), mean_ra = mean(tau_ra))
c(var_dm = var(tau_dm), var_ra = var(tau_ra))
```

5.6 Summary

- Potential outcomes define causal effects, while consistency links the observed outcome to the potential outcome selected by treatment.
- Random assignment identifies the ATE as a difference in observable means.
- The difference in means is exactly the coefficient on treatment in a regression with an intercept and treatment only.
- Baseline covariates are unnecessary for identification in an RCT, but a centered, fully interacted regression adjustment can improve precision without assuming a correct linear outcome model.
- The main assumptions in this chapter are SUTVA, i.i.d. sampling, randomized treatment, suitable moments, and nonsingularity of the covariate second-moment matrix. Natural or quasi-experiments may sometimes justify random assignment, but that claim must be defended in each application.

Chapter 6

Causal Forests

Outline

The ATE summarizes an average effect, but policy questions often concern heterogeneity: who benefits, who is harmed, and how does the effect vary with observed characteristics? This chapter introduces the conditional average treatment effect (CATE), reviews why conventional nonparametric regression becomes difficult in many dimensions, and explains the ideas behind honest causal trees and causal forests.

6.1 CATE and treatment-effect heterogeneity

For a binary treatment, the average treatment effect is

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)].$$

It conceals heterogeneity. The individual treatment effect $Y_i(1) - Y_i(0)$ contains all such variation, but it cannot normally be observed and does not by itself explain which characteristics generate the variation.

Let W_i be a vector of pre-treatment characteristics. The *conditional average treatment effect* is

$$\tau(w) \equiv \mathrm{CATE}(w) = \mathbb{E}[Y_i(1) - Y_i(0) \mid W_i = w]. \tag{6.1}$$

This object organizes heterogeneity by observed characteristics. If W_i is a gender indicator, for example, comparing $\tau(1)$ and $\tau(0)$ describes how the effect differs between those groups. The law of iterated expectations also gives

$$\tau = \mathbb{E}[\tau(W_i)]. \tag{6.2}$$

Thus the CATE function contains the information needed to recover the ATE.

6.1.1 Unconfoundedness, overlap, and identification

Randomized treatment is not necessary if treatment is as good as random after conditioning on observed covariates. The key assumption is

$$D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0)\big) \mid W_i, \tag{6.3}$$

called *unconfoundedness*, *conditional exchangeability*, or *selection on observables*. We also need overlap at the covariate value of interest:

$$0 < \Pr(D_i = 1 \mid W_i = w) < 1. \tag{6.4}$$

Without overlap, one of the two conditional outcome means cannot be learned from the data at that w.

Complete randomization of D_i independently of potential outcomes and W_i implies (6.3), but the converse need not hold. Stratified experiments are a useful example: treatment probabilities may differ by gender, age, or site, while assignment remains random within each stratum. In observational data, (6.3) requires all common causes of treatment and the potential outcomes to be included in W_i. It is a substantive assumption, not something that can be verified from the observed data alone.

For intuition, suppose $Y_i(d) = g_d(W_i, U_i)$. A sufficient condition is that, conditional on W_i, treatment is independent of the outcome-relevant unobservable U_i. Merely allowing treatment to remain related to such an U_i after conditioning on W_i would generally violate unconfoundedness.

Under (6.3) and (6.4),

$$\begin{aligned}
\tau(w) &= \mathbb{E}[Y_i(1) \mid W_i = w] - \mathbb{E}[Y_i(0) \mid W_i = w] \\
&= \mathbb{E}[Y_i(1) \mid D_i = 1, W_i = w] - \mathbb{E}[Y_i(0) \mid D_i = 0, W_i = w] \\
&= \mathbb{E}[Y_i \mid D_i = 1, W_i = w] - \mathbb{E}[Y_i \mid D_i = 0, W_i = w]. \qquad (6.5)
\end{aligned}$$

The second equality uses unconfoundedness and the last uses consistency. Define

$$\mu_d(w) = \mathbb{E}[Y_i \mid D_i = d, W_i = w].$$

Estimating the CATE is now a problem of estimating the two regression functions μ_1 and μ_0, or estimating their contrast directly.

If W is discrete, a cell mean may work. If W is continuous, the event $W = w$ usually has probability zero, so exact matching is unavailable. We need a method that borrows information across nearby or otherwise similar observations.

6.2 Traditional nonparametric methods

There are two broad ways to estimate an unknown regression function.

6.2.1 Local methods

A local method estimates $\mu_d(w)$ using observations with W_i near w. With a scalar W, a simple estimator averages observations for which $|W_i - w| \le h$. The bandwidth h controls a familiar trade-off:

- a larger h uses more observations and reduces variance, but introduces more approximation bias;
- a smaller h focuses more tightly on w and reduces bias, but increases variance.

If W has dimension p and each coordinate uses a bandwidth of order h, the effective sample size is of order nh^p. Consequently, a typical pointwise standard error is of order

$$\frac{1}{\sqrt{nh^p}}, \qquad (6.6)$$

not $\sqrt{nh^p}$. As p grows, it becomes difficult to find observations close to w in every coordinate. This is the *curse of dimensionality*.

6.2.2 Global methods

A global method approximates the entire function. For known basis functions $b_1, b_2, \ldots$, one may write

$$\mu_d(w) \approx \sum_{j=1}^{J} a_{dj} b_j(w).$$

Polynomials, B-splines, and wavelets are common choices. Increasing J reduces approximation bias but requires estimating more coefficients. With several covariates, the number of basis terms needed for a flexible approximation can grow quickly, so the curse of dimensionality remains.

Many machine-learning methods fit within this local–global distinction. Trees and forests create adaptive neighborhoods and can be viewed as adaptive local methods. Neural networks, studied later, construct a global approximation.

6.3 Regression trees, causal trees, and causal forests

The central insight of a tree is that geometric closeness is sufficient but not necessary for similar regression values. Two distant covariate vectors may have the same $\mu_d(w)$, while a function may change sharply over a small distance. A tree uses the outcomes to learn rectangular regions within which the target is approximately constant.

6.3.1 A regression-tree sketch

Suppose $W = (W_1, W_2)'$ lies in the unit square. A simplified recursive partitioning algorithm proceeds as follows.

1. For each covariate W_j and each admissible threshold t, split the current node into $W_j \leq t$ and $W_j > t$.
2. Evaluate how much the split reduces within-node squared error, equivalently how strongly the two child-node means differ after accounting for their sizes.
3. Choose the best admissible split and repeat within each child node.
4. Stop when a depth or minimum-leaf-size rule is reached. Predict at w by averaging outcomes in the terminal leaf containing w.

The one-coordinate-at-a-time splits make computation manageable and give an interpretable binary-tree representation. In practice, candidate thresholds are the finitely many cut points induced by the sample, and split criteria include regularization to avoid tiny or severely imbalanced leaves.

In Figure 6.1, the top-left partition is not generally representable by recursive one-coordinate splits. The top-right partition is; the bottom-left panel records the corresponding decision rules, and the bottom-right panel shows the fitted step function.

6.3.2 From a regression tree to a causal tree

A causal tree targets the contrast in (6.5), not merely an outcome mean. In a simple description, each proposed child node contains a treated mean and a control mean, and hence an estimated treatment effect. A useful split is one for which the treatment-effect estimates differ substantially across the two children, subject to penalties for estimation noise and constraints ensuring that both treatment arms are represented.

This description conveys the intuition, but production implementations use more carefully regularized splitting objectives than the raw squared difference between two noisy effect estimates.

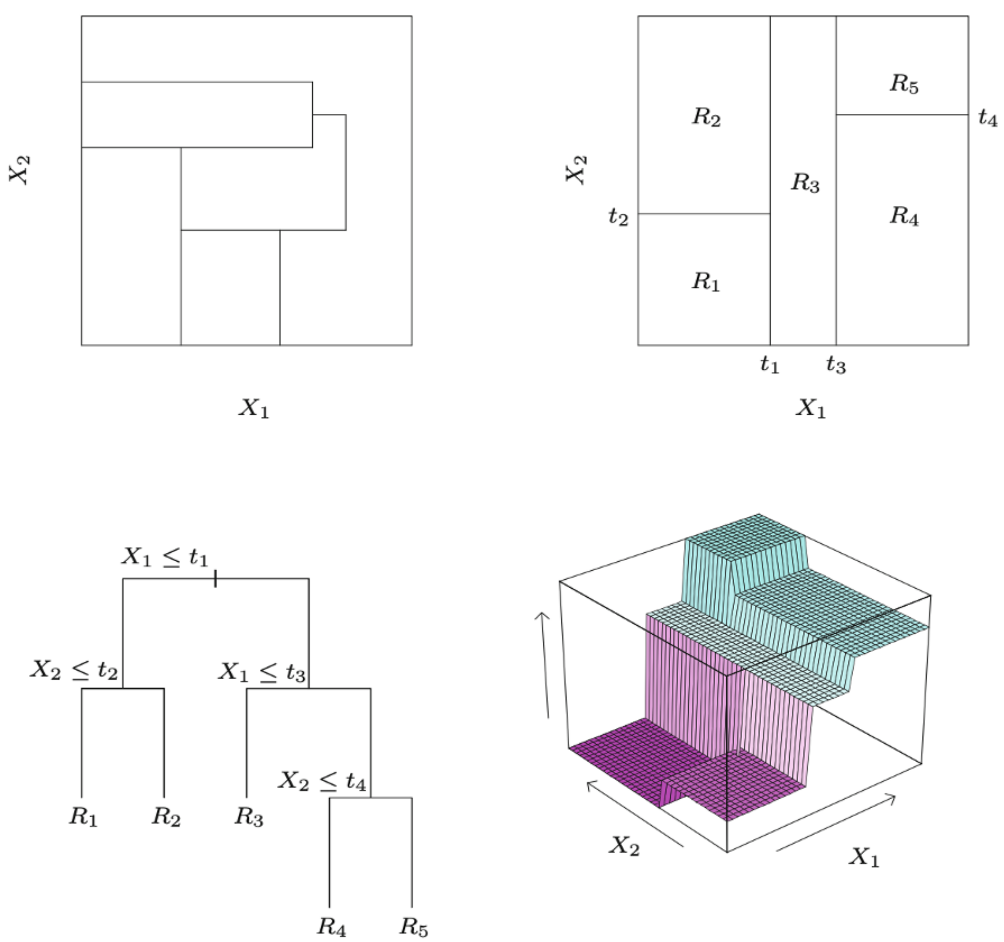

Figure 6.1: A generic partition, a recursive rectangular partition, its tree representation, and the resulting piecewise-constant fit. Adapted from Friedman, Hastie, and Tibshirani, *The Elements of Statistical Learning* (2009), p. 306.

6.3.3 Honesty

If the same outcomes select the partition and estimate effects inside the selected leaves, the procedure can chase noise. Conditional on the selected leaf, the within-leaf outcomes are no longer an ordinary, exogenously selected sample. *Honesty* separates the two tasks:

1. randomly divide a subsample into a splitting sample and an estimation sample;
2. use only the splitting sample to construct the tree;
3. hold the partition fixed and use only the estimation sample to estimate effects inside its leaves.

Under suitable regularity conditions—including overlap, regular splits, growing leaf information, and smoothness—honest tree and forest estimators can be shown to be consistent and asymptotically normal. These are asymptotic statements, not a claim of exact finite-sample unbiasedness. Pointwise nonparametric rates are typically slower than the parametric $n^{-1/2}$ rate.

6.3.4 Causal forests

A single causal tree is unstable: a modest change in the data can alter an early split and therefore the entire partition. A causal forest averages many randomized honest trees:

1. draw a subsample of the observations;
2. grow an honest causal tree, usually considering a random subset of covariates at each split;
3. repeat for B trees;
4. average the B tree predictions at the target w.

Subsampling and random covariate selection make the trees less correlated; averaging then stabilizes the prediction. Across trees, an observation may be used for splitting in some trees, effect estimation in others, and omitted from still others.

Three tuning ideas are especially important:

Covariate subsampling. Considering only a random subset of the p covariates at a split reduces computation and decorrelates the trees. The `grf` package chooses sensible defaults that can be tuned.

Minimum leaf size. Large leaves can smooth over real heterogeneity; tiny leaves have noisy treated and control means.

Balanced splits. Each child needs enough observations, including enough treated and control observations, for stable effect estimation and overlap.

6.3.5 Why ordinary linear regression is not enough

Under complete randomization, the treatment coefficient in a short regression identifies the ATE even if the outcome model is nonlinear. CATE is different: it is a function of W. A linear interaction model such as

$$Y = \alpha + D\tau + W'\beta + DW'\delta + u$$

restricts the CATE to $\tau + W'\delta$. If the true CATE is nonlinear, this function is misspecified. Moreover, under unconfoundedness D may be related to W, so omitted nonlinear functions of

W can also confound treatment. Causal forests allow flexible nonlinearities and interactions, though they do not remove the need for unconfoundedness and overlap.

6.4 Implementation with `grf`

The next simulation has $n = 2{,}000$, ten covariates, and a treatment probability that changes with W_1. The outcome model is

$$Y = \max\{W_1, 0\}D + W_2 + \min\{W_3, 0\} + \varepsilon,$$

so the true CATE is $\tau(w) = \max\{w_1, 0\}$.

```
library(grf)
library(DiagrammeR)

set.seed(5280)
n <- 2000
p <- 10
W <- matrix(rnorm(n * p), nrow = n, ncol = p)
D <- rbinom(n, 1, 0.4 + 0.2 * (W[, 1] > 0))
Y <- pmax(W[, 1], 0) * D + W[, 2] + pmin(W[, 3], 0) + rnorm(n)

W_test <- matrix(0, nrow = 101, ncol = p)
W_test[, 1] <- seq(-2, 2, length.out = 101)

# A quick exploratory forest and one of its trees
forest_quick <- causal_forest(W, Y, D)
plot(get_tree(forest_quick, 1))

# A larger forest for final estimates and variance estimates
forest_full <- causal_forest(W, Y, D, num.trees = 4000)
pred <- predict(forest_full, W_test, estimate.variance = TRUE)
se <- sqrt(pred$variance.estimates)

plot(W_test[, 1], pred$predictions,
     ylim = range(pred$predictions - 1.96 * se,
                  pred$predictions + 1.96 * se, 0, 2),
     xlab = expression(W[1]), ylab = "CATE", type = "l")
lines(W_test[, 1], pred$predictions + 1.96 * se, lty = 2)
lines(W_test[, 1], pred$predictions - 1.96 * se, lty = 2)
lines(W_test[, 1], pmax(W_test[, 1], 0), col = 2)
legend("topleft", c("forest", "95% pointwise interval", "truth"),
       lty = c(1, 2, 1), col = c(1, 1, 2), bty = "n")
```

Four thousand trees reduce Monte Carlo noise in variance estimates but can be slow in a browser. The live companion uses fewer trees for a quick classroom demonstration; the substantive workflow should use more trees and check that results are stable as `num.trees` increases.

We can also revisit the survey-wording experiment from Section 5.4.2. Here the test grid fixes age at 30 and varies education.

```
gss <- read.csv("data/welfare-small.csv")
Y <- as.numeric(gss$Y)
D <- as.numeric(gss$D)
W <- as.matrix(gss[, c("age", "educ")])
```

```
W_test <- cbind(age = 30,
                educ = seq(min(gss$educ), max(gss$educ), by = 1))

forest_gss <- causal_forest(W, Y, D, num.trees = 4000)
pred_gss <- predict(forest_gss, W_test, estimate.variance = TRUE)
se_gss <- sqrt(pred_gss$variance.estimates)

plot(W_test[, "educ"], pred_gss$predictions,
     ylim = range(pred_gss$predictions - 1.96 * se_gss,
                  pred_gss$predictions + 1.96 * se_gss),
     xlab = "Education", ylab = "Estimated CATE", type = "l")
lines(W_test[, "educ"], pred_gss$predictions + 1.96 * se_gss, lty = 2)
lines(W_test[, "educ"], pred_gss$predictions - 1.96 * se_gss, lty = 2)
```

The confidence limits above are pointwise. They should not be interpreted as a simultaneous confidence band for the whole CATE curve. In an observational application, predictive fit also cannot validate unconfoundedness.

6.5 Summary

- The CATE organizes treatment-effect heterogeneity by pre-treatment characteristics, and averaging it recovers the ATE.
- Unconfoundedness and overlap identify the CATE as a contrast of two observable conditional means.
- Conventional nonparametric methods face the curse of dimensionality; a local estimator's standard error is typically of order $(nh^p)^{-1/2}$.
- Causal trees learn adaptive regions targeted to treatment-effect heterogeneity. Honesty separates partition selection from effect estimation.
- Causal forests average many randomized honest trees, improving stability and permitting pointwise inference under regularity conditions.
- Flexible algorithms do not make causal assumptions disappear: credible unconfoundedness, overlap, SUTVA, and a clearly defined target population remain essential.

Chapter 7

Doubly Robust Learning

Outline

Chapter 6 identified conditional treatment effects under unconfoundedness. We now return to the average treatment effect (ATE). Ordinary least squares need not recover the ATE when treatment is only conditionally random. We develop outcome-regression and inverse-propensity-weighting representations, then combine them in the augmented inverse propensity weighted (AIPW) score. Cross-fitting makes this score compatible with flexible machine-learning estimates of nuisance functions.

7.1 ATE under unconfoundedness and the failure of OLS

Let W_i contain pre-treatment covariates. We maintain consistency, i.i.d. sampling, unconfoundedness,

$$D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0)\big) \mid W_i, \tag{7.1}$$

and overlap,

$$0 < e(W_i) < 1 \quad \text{almost surely}, \qquad e(w) = \Pr(D_i = 1 \mid W_i = w). \tag{7.2}$$

Define the outcome regressions

$$\mu_d(w) = \mathbb{E}[Y_i \mid D_i = d, W_i = w], \qquad d \in \{0, 1\}. \tag{7.3}$$

By the CATE identification argument,

$$\tau \equiv \text{ATE} = \mathbb{E}[\mu_1(W_i) - \mu_0(W_i)]. \tag{7.4}$$

This is often called the outcome-regression formula, the g-formula, or standardization.

Identification does not imply that an arbitrary linear regression is correctly specified. Consider the following data-generating process. Treatment depends on W, so it is not completely randomized, but the independent variable V makes treatment random conditional on W. The individual effect is $1 + W^2$, hence the ATE is $1 + \mathbb{E}[W^2] = 2$.

```r
library(lmtest)
library(sandwich)

set.seed(5280)
n <- 5000
W <- rnorm(n)
V <- rnorm(n)
D <- as.integer(W + V > 0)
epsilon <- rnorm(n)
Y <- 1 + D + W + D * W^2 + sin(W) + epsilon

fit_short <- lm(Y ~ D)
fit_linear <- lm(Y ~ D + W)
coeftest(fit_short, vcov. = vcovHC(fit_short, type = "HC2"))["D", ]
coeftest(fit_linear, vcov. = vcovHC(fit_linear, type = "HC2"))["D", ]
```

Both regressions omit nonlinear terms that are related to treatment. Their treatment coefficients are therefore not generally the ATE. Adding a sufficiently rich and correctly specified outcome model could solve the problem, but we would like a method that accommodates flexible estimation and offers protection against one nuisance model being misspecified.

7.2 The propensity score and two representations of the ATE

The function $e(W_i)$ in (7.2) is the *propensity score*. It summarizes how observed covariates predict treatment. Under complete randomization it is constant; under conditional randomization it may vary with W_i.

7.2.1 The balancing-score result

Rosenbaum and Rubin's balancing-score theorem states that

$$D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0)\big) \mid e(W_i) \tag{7.5}$$

whenever (7.1) holds. To see the key argument, let $Z_i = (Y_i(1), Y_i(0))$. Since D_i is binary,

$$\begin{aligned}
\Pr(D_i = 1 \mid Z_i, e(W_i)) &= \mathbb{E}[D_i \mid Z_i, e(W_i)] \\
&= \mathbb{E}[\mathbb{E}[D_i \mid Z_i, W_i] \mid Z_i, e(W_i)] \\
&= \mathbb{E}[e(W_i) \mid Z_i, e(W_i)] = e(W_i),
\end{aligned}$$

where unconfoundedness gives the third line. Likewise, $\Pr(D_i = 1 \mid e(W_i)) = e(W_i)$. Thus the conditional treatment probability does not depend on Z_i once the propensity score is held fixed.

The theorem makes a high-dimensional conditioning problem scalar, but it does not mean that one can ignore W when estimating the propensity score. Misspecifying $e(W)$ can still invalidate a propensity-score analysis.

7.2.2 Inverse propensity weighting

The ATE also has the representation

$$\tau = \mathbb{E}\left[\frac{D_i Y_i}{e(W_i)} - \frac{(1 - D_i) Y_i}{1 - e(W_i)}\right]. \tag{7.6}$$

For the treated component, consistency and iterated expectations give

$$\begin{aligned}
\mathbb{E}\left[\frac{D_i Y_i}{e(W_i)}\right] &= \mathbb{E}\left[\frac{D_i Y_i(1)}{e(W_i)}\right] \\
&= \mathbb{E}\left[\frac{\mathbb{E}[D_i Y_i(1) \mid W_i]}{e(W_i)}\right] \\
&= \mathbb{E}\left[\frac{\mathbb{E}[D_i \mid W_i]\mathbb{E}[Y_i(1) \mid W_i]}{e(W_i)}\right] \\
&= \mathbb{E}[Y_i(1)].
\end{aligned}$$

The third line uses unconfoundedness. The control component similarly equals $\mathbb{E}[Y_i(0)]$, proving (7.6). Overlap is needed because the denominators must be defined; near-zero denominators also explain why weak overlap can make IPW unstable.

We now have two identification formulas: outcome regression in (7.4) and IPW in (7.6). AIPW combines them.

7.3 The AIPW score and double robustness

For generic functions m_1, m_0, and q taking values strictly between zero and one, define

$$\psi(O_i; m_1, m_0, q) = m_1(W_i) - m_0(W_i) + \frac{D_i\{Y_i - m_1(W_i)\}}{q(W_i)} - \frac{(1-D_i)\{Y_i - m_0(W_i)\}}{1-q(W_i)}, \tag{7.7}$$

where $O_i = (Y_i, D_i, W_i)$. If the true nuisance functions $\eta_0 = (\mu_1, \mu_0, e)$ were known, the oracle estimator would be

$$\widehat{\tau}^*_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n} \psi(O_i; \mu_1, \mu_0, e). \tag{7.8}$$

The augmentation terms have conditional mean zero. For example,

$$\mathbb{E}\left[\frac{D_i\{Y_i - \mu_1(W_i)\}}{e(W_i)} \;\Big|\; W_i\right] = 0.$$

Hence $\mathbb{E}[\psi(O_i; \eta_0)] = \tau$. The score can also be rearranged as

$$\begin{aligned}\psi(O_i; m_1, m_0, q) &= \frac{D_i Y_i}{q(W_i)} - \frac{(1-D_i)Y_i}{1-q(W_i)} \\ &\quad + m_1(W_i)\left\{1 - \frac{D_i}{q(W_i)}\right\} - m_0(W_i)\left\{1 - \frac{1-D_i}{1-q(W_i)}\right\}.\end{aligned} \tag{7.9}$$

These two forms reveal *double robustness*:

- If $m_d = \mu_d$ for $d = 0, 1$, then the residual augmentation terms in (7.7) have conditional mean zero for any admissible q. Therefore $\mathbb{E}[\psi] = \mathbb{E}[\mu_1(W) - \mu_0(W)] = \tau$.
- If $q = e$, then

  $$\mathbb{E}\left[1 - \frac{D}{e(W)} \;\Big|\; W\right] = 0, \qquad \mathbb{E}\left[1 - \frac{1-D}{1-e(W)} \;\Big|\; W\right] = 0.$$

  Consequently, the last line of (7.9) has mean zero for any integrable m_1, m_0, while its first line identifies the ATE.

The conditioning on W in the second argument is essential: an unconditional mean-zero equality alone would not justify multiplication by an arbitrary function of W.

Let $\widehat{\mu}_d$ and $\widehat{e}$ be estimated nuisance functions. The plug-in AIPW estimator is

$$\widehat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n} \psi(O_i; \widehat{\mu}_1, \widehat{\mu}_0, \widehat{e}). \tag{7.10}$$

Subject to regularity conditions, it remains consistent when either both outcome regressions are consistently estimated or the propensity score is consistently estimated. Double robustness protects against one class of nuisance-model misspecification; it does not protect against violations of unconfoundedness, overlap, consistency, or sampling assumptions.

7.4 Cross-fitting and double machine learning

Flexible machine-learning estimates can overfit. Evaluating a nuisance function on the same observations used to train it creates dependence that complicates the first-order expansion of (7.10). *Cross-fitting* removes this own-observation link.

Partition the sample into K folds $I_1, \ldots, I_K$. For each fold k:

1. estimate μ_1, μ_0, and e using observations outside I_k; call the resulting functions $\widehat{\mu}_1^{(-k)}, \widehat{\mu}_0^{(-k)}, \widehat{e}^{(-k)}$;
2. evaluate those functions for observations $i \in I_k$ and compute their AIPW scores.

The cross-fitted estimator is

$$\widehat{\tau}_{\mathrm{DML}} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \psi\left(O_i; widehat\mu_1^{(-k)}, \widehat{\mu}_0^{(-k)}, \widehat{e}^{(-k)}\right). \tag{7.11}$$

Every observation is used for score evaluation once and for nuisance training in the other folds.

Why can slowly estimated nuisance functions still yield root-n inference for τ? The AIPW score is orthogonal: its population expectation is locally insensitive, to first order, to small perturbations of the nuisance functions. The leading remainder is governed by a *product* of nuisance errors. A typical sufficient condition is

$$\|\widehat{e} - e\|_2 \left(\|\widehat{\mu}_1 - \mu_1\|_2 + \|\widehat{\mu}_0 - \mu_0\|_2\right) = o_p(n^{-1/2}), \tag{7.12}$$

together with consistent nuisance estimates, bounded moments, and strong overlap: $e(W)$ is bounded away from zero and one.

If the propensity and outcome-regression mean-squared errors are respectively of orders n^{-a} and n^{-b}, their L^2 errors have orders $n^{-a/2}$ and $n^{-b/2}$. Thus (7.12) corresponds to $a + b > 1$ when these are exact big-O rates; an equality case needs a stronger little-o statement. It is incorrect simply to compare nuisance variances with $1/n$ without taking square roots and the product remainder into account.

Under appropriate conditions,

$$\sqrt{n}(\widehat{\tau}_{\mathrm{DML}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{\mathrm{eff}}), \qquad V_{\mathrm{eff}} = \mathrm{Var}\{\psi(O_i; \eta_0)\}. \tag{7.13}$$

The centered score $\psi(O_i; \eta_0) - \tau$ is the efficient influence function for the ATE in the nonparametric unconfoundedness model. Consequently, an estimator with the expansion in (7.13) attains the semiparametric efficiency bound among regular estimators in that model. This is a precise efficiency claim, not a statement that AIPW is unconditionally "best" in every finite sample.

An estimated standard error is obtained from the sample variance of the cross-fitted scores:

$$\widehat{\mathrm{se}}(\widehat{\tau}_{\mathrm{DML}}) = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^{n} (\widehat{\psi}_i - \widehat{\tau}_{\mathrm{DML}})^2}. \tag{7.14}$$

The familiar Wald confidence interval then applies when the asymptotic conditions are credible.

7.5 Implementation with generalized random forests

The `grf` package estimates nuisance components internally and uses out-of-bag predictions to construct an orthogonalized ATE estimate. We return to the simulation from Chapter 6.

```r
library(grf)

set.seed(5280)
n <- 2000
p <- 10
W <- matrix(rnorm(n * p), nrow = n, ncol = p)
```

```
D <- rbinom(n, 1, 0.4 + 0.2 * (W[, 1] > 0))
Y <- pmax(W[, 1], 0) * D + W[, 2] + pmin(W[, 3], 0) + rnorm(n)

# Use many trees for stable final inference.
forest <- causal_forest(W, Y, D, num.trees = 4000)
average_treatment_effect(
  forest,
  target.sample = "all",
  method = "AIPW"
)
```

Here the true ATE is $\mathbb{E}[\max\{W_1, 0\}] = 1/\sqrt{2\pi} \approx 0.399$. Comparing the estimate with this truth is a useful simulation check. Four thousand trees are appropriate for a substantive analysis but unnecessarily slow for a short browser demonstration; the live companion uses a smaller forest and labels that choice explicitly.

In applied work, also inspect the estimated propensity scores, sensitivity to forest tuning, and the effective target population. Extreme propensities can make AIPW noisy despite its robustness properties. If overlap is poor, changing the estimand, trimming by a pre-specified rule, or collecting better comparison data may be more defensible than relying on a few enormous weights.

7.6 Summary

- Under unconfoundedness and overlap, the ATE has both an outcome-regression representation and an inverse-propensity-weighted representation.
- OLS with an arbitrary linear specification need not estimate the ATE when treatment depends on covariates.
- AIPW combines the two representations and is consistent if either the two outcome regressions or the propensity score is consistently estimated, subject to regularity conditions.
- Cross-fitting permits flexible nuisance learners while avoiding own-observation overfitting in the score.
- Orthogonality makes the leading bias depend on the product of nuisance errors. With strong overlap and suitable product rates, AIPW is root-n normal and semiparametrically efficient.
- These statistical protections do not repair a failure of the identifying causal assumptions.

Chapter 8

Sharp Regression Discontinuity Designs

Motivation. What can we learn when treatment is a deterministic function of a running variable?

Identification. Why only the treatment effect at the cutoff is identified without additional assumptions.

Estimation. How local linear regression estimates the two limits.

Application. Incumbency effects in U.S. House elections.

8.1 A deterministic treatment rule

In earlier chapters, unconfoundedness was paired with overlap. For a scalar covariate W_i, those assumptions allowed us to compare treated and untreated units at the same value $W_i = w$. A sharp regression discontinuity (RD) design is deliberately different. Treatment obeys a deterministic rule,

$$D_i = \mathbf{1}\{W_i \geq c\}, \tag{8.1}$$

where c is a known cutoff. Conditional on W_i, treatment is degenerate; there is no treated–untreated overlap except in a limiting sense at the cutoff. Thus the usual unconfoundedness formula

$$\mathbb{E}[Y_i \mid D_i = 1, W_i = w] - \mathbb{E}[Y_i \mid D_i = 0, W_i = w]$$

is not well-defined: below c the first conditional mean is unavailable, and above c the second is unavailable.

Write the observed outcome as

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The conditional average treatment effect is

$$\tau(w) = \mathbb{E}[Y_i(1) - Y_i(0) \mid W_i = w]. \tag{8.2}$$

Away from the cutoff, one of its two components is missing:

$$\tau(w) = \begin{cases} \mathbb{E}[Y_i \mid W_i = w] - \mathbb{E}[Y_i(0) \mid W_i = w], & w \geq c, \\ \mathbb{E}[Y_i(1) \mid W_i = w] - \mathbb{E}[Y_i \mid W_i = w], & w < c. \end{cases}$$

Without extrapolation, therefore, we cannot identify the entire function $\tau(\cdot)$ or the population ATE.

8.2 Identification at the cutoff

The cutoff is special because observations can approach it from either side. Assume that

$$m_d(w) = \mathbb{E}[Y_i(d) \mid W_i = w], \qquad d \in \{0,1\},$$

is continuous at c. Equation (8.1) and consistency imply

$$\lim_{w \downarrow c} \mathbb{E}[Y_i \mid W_i = w] = m_1(c),$$
$$\lim_{w \uparrow c} \mathbb{E}[Y_i \mid W_i = w] = m_0(c).$$

Here $w \downarrow c$ approaches from above, whereas $w \uparrow c$ approaches from below. Subtracting the two limits identifies

$$\tau_{RD} \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid W_i = c] = \lim_{w \downarrow c} \mathbb{E}[Y_i \mid W_i = w] - \lim_{w \uparrow c} \mathbb{E}[Y_i \mid W_i = w]. \tag{8.3}$$

Important

RD identification comes from the deterministic assignment rule and continuity of the untreated and treated conditional means at the cutoff. It does not come from the usual unconfoundedness-plus-overlap argument.

The continuity requirement says that, absent the change in treatment status, the expected outcome would not jump exactly at c. This is plausible when units just above and just below the threshold are otherwise comparable. It is not automatic. Other programs changing at the same cutoff, precise manipulation of W_i, or a discontinuous change in the composition of units can invalidate the design.

The estimand is local. An RD study of a program with an income threshold $c = \$1{,}000$, for example, estimates the effect for families at that threshold. It does not by itself identify the effect for families earning \$500. Whether τ_{RD} is substantively interesting is therefore a design question, not a statistical one.

8.3 Local linear estimation

Equation (8.3) asks for two boundary values of a conditional mean. Near the cutoff, approximate that mean by a different line on each side:

$$Y_i = \beta_0^+ + \beta_1^+(W_i - c) + \varepsilon_i^+, \qquad W_i \geq c, \tag{8.4}$$
$$Y_i = \beta_0^- + \beta_1^-(W_i - c) + \varepsilon_i^-, \qquad W_i < c. \tag{8.5}$$

Then $\beta_0^+ - \beta_0^-$ approximates the discontinuity. Equivalently, fit one interacted regression,

$$Y_i = \gamma_0 + \gamma_1 D_i + \gamma_2(W_i - c) + \gamma_3 D_i(W_i - c) + \varepsilon_i, \tag{8.6}$$

using only observations near c. The mapping is

$$\gamma_0 = \beta_0^-, \quad \gamma_2 = \beta_1^-, \quad \gamma_0 + \gamma_1 = \beta_0^+, \quad \gamma_2 + \gamma_3 = \beta_1^+,$$

so γ_1 is the estimated jump.

Let

$$X_i = \big(1, D_i, W_i - c, D_i(W_i - c)\big)', \qquad K_{hi} = K\left(\frac{W_i - c}{h}\right).$$

A kernel K gives greater weight to observations closer to the cutoff, and $h > 0$ is the bandwidth. Weighted least squares solves

$$\frac{1}{n}\sum_{i=1}^{n} K_{hi} X_i (Y_i - X_i' \widehat{\gamma}) = 0, \tag{8.7}$$

and hence

$$\widehat{\gamma} = \left(\frac{1}{n}\sum_i K_{hi} X_i X_i'\right)^{-1} \left(\frac{1}{n}\sum_i K_{hi} X_i Y_i\right). \tag{8.8}$$

The RD point estimate is the second element, $\widehat{\gamma}_1$.

As n grows, the bandwidth must shrink so the linear approximation becomes local, but not so quickly that too few observations remain. This creates the usual bias–variance tradeoff:

- a small h reduces approximation bias but increases variance;
- a large h uses more data but risks fitting observations for which a straight line is a poor approximation.

At an MSE-optimal bandwidth, the leading smoothing bias need not be negligible for inference. Modern practice therefore uses robust bias correction and reports sensitivity to bandwidth and polynomial-order choices. The older IK bandwidth of Imbens and Kalyanaraman is useful historically, but it is not the only or necessarily preferred modern choice.

8.3.1 Covariates

Additional pre-treatment covariates are not required for the identification argument in Section 8.2. They may improve precision and can help diagnose whether units look comparable near the cutoff. They should be included in a way that preserves the local RD specification; adding post-treatment covariates can instead change or destroy the causal estimand.

8.3.2 Why not a high-order global polynomial?

It is possible to fit separate polynomials over the entire range of W_i. Such fits were common in early applications, including the descriptive Lee figure below. High-order global polynomials can be unstable near a boundary and can give distant observations excessive influence. Local linear or local quadratic methods are normally safer for primary estimation.

8.4 Application: electoral incumbency

Lee (2008) studies whether winning a U.S. House election raises the Democratic candidate's vote share in the next election. Define

- the running variable as the Democratic vote-share margin in election t, recentered so zero means a tie; and
- the outcome as the Democratic vote share in election $t + 1$.

Winning election t is endogenous in the full sample: stronger candidates and districts both win more often and perform better later. Very close elections, however, compare districts on opposite sides of an institutional threshold. The discontinuity at zero in Figure 8.1 is interpreted as the incumbency effect under the continuity assumptions above.

Other well-known applications study mayoral wages and performance, colonial institutions and development, financial-aid offers and college enrollment, and angel funding and startup

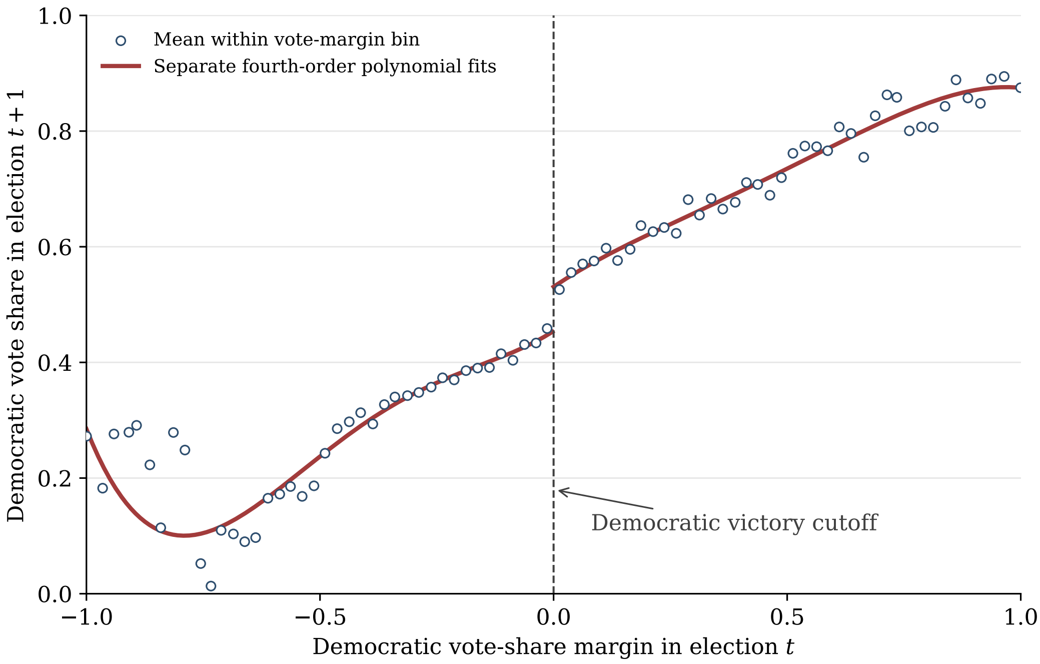

Figure 8.1: A Lee-style electoral RD illustration. Circles are equal-width-bin means from the course data; curves are separate fourth-order polynomial fits used for visualization. Primary estimation should use a local method.

growth. In every case, the central task is to defend the institutional cutoff and the continuity of potential outcomes there.

8.5 R implementation

The following small simulation uses the `rdd` package. Its default `RDestimate()` call uses a triangular kernel and an automatically chosen bandwidth. The companion Quarto Live page runs the same code in the browser.

```
library(rdd)

set.seed(5280)
x <- runif(1000, -1, 1)
y <- 3 + 2 * x + 10 * (x >= 0) + rnorm(1000)

fit_auto <- RDestimate(y ~ x)
fit_grid <- RDestimate(y ~ x,
                       bw = c(0.10, 0.20, 0.30),
                       kernel = "gaussian")
summary(fit_auto)
summary(fit_grid)
plot(fit_auto)
```

The Lee-style data can be loaded as follows. The file path in the live page is relative to the chapter file.

```
library(rdd)
house <- read.csv("house.csv")
lee_local <- RDestimate(y ~ x, data = house)
```

```
summary(lee_local)
plot(lee_local)
```

Summary

- Sharp RD assigns treatment deterministically at a known cutoff, so ordinary overlap fails.
- Continuity identifies one local effect: the jump in the conditional mean of the outcome at the cutoff.
- Local linear regression estimates separate boundary values and avoids relying on a high-order global polynomial.
- Bandwidth choice, bias correction, manipulation, concurrent policies, and the local interpretation are substantive parts of the design.

Chapter 9

Instrumental Variables and Local Average Treatment Effects

Motivation. What can a randomized offer identify when people do not fully comply with their assignment?

LATE. Define the causal effect for compliers and state the assumptions that identify it.

IV estimation. Derive the Wald estimand and its moment-equation representation.

Applications. Discuss where credible instruments come from and why exclusion and relevance require substantive arguments.

9.1 Noncompliance and intent to treat

Suppose access to an HPV vaccine is allocated by lottery. Let $Z_i = 1$ mean that individual i wins a free offer and let $D_i = 1$ mean that she actually receives the vaccine. Even if Z_i is randomized, compliance need not be perfect:

- some winners may decline the vaccine; and
- some nonwinners may obtain it elsewhere.

The difference

$$\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]$$

is the *intent-to-treat* (ITT) effect of the offer on the outcome. It is not generally the average effect of receiving the vaccine.

To describe noncompliance, define the potential treatment $D_i(z)$: whether unit i would receive treatment if the instrument were set to z. Only one of $D_i(0)$ and $D_i(1)$ is observed, because

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0). \tag{9.1}$$

The pair $(D_i(0), D_i(1))$ defines four principal strata.

We never observe an individual's type directly. Calling a unit a complier is also application-specific: it depends on which direction of the instrument is defined as encouragement.

Definition 9.1 (Local average treatment effect)**.** For binary Z_i and D_i, the local average treatment effect is

$$LATE = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]. \tag{9.2}$$

Table 9.1: Compliance types for a binary instrument and treatment

Type	$D_i(0)$	$D_i(1)$
Never-taker	0	0
Complier	0	1
Defier	1	0
Always-taker	1	1

It is the average treatment effect for compliers.

If everyone is a complier, LATE equals the population ATE. Otherwise it is a local parameter for the subpopulation whose treatment status is changed by the instrument. Different valid instruments may therefore identify different LATEs.

9.2 Identification of LATE

It helps to separate four assumptions that are sometimes bundled together.

1. **Random assignment (instrument exogeneity):**
$$Z_i \perp\!\!\!\perp \big(Y_i(0), Y_i(1), D_i(0), D_i(1)\big). \tag{9.3}$$

2. **Exclusion:** the instrument affects the outcome only through treatment. If outcomes were initially written $Y_i(d,z)$, exclusion says $Y_i(d,z) = Y_i(d)$ for every d, z.

3. **Relevance:**
$$\mathbb{E}[D_i \mid Z_i = 1] \neq \mathbb{E}[D_i \mid Z_i = 0]. \tag{9.4}$$

4. **Monotonicity:**
$$D_i(1) \geq D_i(0) \quad \text{almost surely.} \tag{9.5}$$
After orienting Z_i as an encouragement, there are no defiers.

Random assignment and exclusion are distinct. A lottery can be randomized but still violate exclusion if winning changes behavior through publicity, income, or expectations even when treatment receipt is held fixed.

Theorem 9.1 (Imbens–Angrist LATE theorem)**.** *Under random assignment, exclusion, relevance, and monotonicity,*
$$LATE = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}. \tag{9.6}$$
The numerator is the outcome ITT and the denominator is the first-stage ITT.

Proof. Consistency and exclusion give $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. Using Equation (9.1) and random assignment,
$$\begin{aligned} &\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] \\ &\quad = \mathbb{E}\left[(D_i(1) - D_i(0))(Y_i(1) - Y_i(0))\right]. \end{aligned}$$
Under monotonicity, $D_i(1) - D_i(0)$ equals one for compliers and zero for always- and never-takers. Hence the last display equals
$$\Pr\{D_i(1) > D_i(0)\}\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)].$$
Similarly,
$$\begin{aligned} \mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0] &= \mathbb{E}[D_i(1) - D_i(0)] \\ &= \Pr\{D_i(1) > D_i(0)\}. \end{aligned}$$
Relevance makes this probability nonzero, so division yields Equation (9.6). □

Monotonicity is substantive, not a mathematical convenience. It can be credible for a one-sided encouragement but implausible if the instrument pushes different people in opposite directions. Without monotonicity, the Wald ratio mixes complier and defier effects with opposite signs.

9.2.1 A linear moment representation

For binary Z_i, define

$$\gamma = \mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0],$$
$$\delta = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0].$$

The linear projections on $(1, Z_i)$ can be written

$$D_i = \gamma_0 + \gamma Z_i + \nu_i, \qquad \mathbb{E}[\nu_i] = \mathbb{E}[Z_i \nu_i] = 0,$$
$$Y_i = \delta_0 + \delta Z_i + \mu_i, \qquad \mathbb{E}[\mu_i] = \mathbb{E}[Z_i \mu_i] = 0.$$

Because relevance implies $\gamma \neq 0$, substituting the first equation into the second produces

$$Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i] = \mathbb{E}[Z_i \varepsilon_i] = 0, \tag{9.7}$$

where $\beta_1 = \delta/\gamma = LATE$. In general $\mathrm{Cov}(D_i, \varepsilon_i) \neq 0$, which is precisely why OLS is not appropriate.

Equation (9.7) is a useful moment representation. It does not require the individual treatment effect to be constant or the conditional mean of Y_i to be globally linear in D_i.

9.2.2 Conditional LATE

Let W_i contain pre-instrument covariates. A conditional design assumes, for example,

$$Z_i \perp\!\!\!\perp \big(Y_i(0), Y_i(1), D_i(0), D_i(1)\big) \mid W_i,$$

together with conditional exclusion, relevance, and monotonicity. One possible structural shorthand is $Y_i = g(D_i, W_i, U_i)$ and $D_i = h(Z_i, W_i, V_i)$ with $Z_i \perp\!\!\!\perp (U_i, V_i) \mid W_i$; importantly, the treatment equation depends on Z_i, not on D_i itself.

At covariate values with a nonzero conditional first stage,

$$CLATE(w) = \frac{\mathbb{E}[Y_i \mid Z_i = 1, W_i = w] - \mathbb{E}[Y_i \mid Z_i = 0, W_i = w]}{\mathbb{E}[D_i \mid Z_i = 1, W_i = w] - \mathbb{E}[D_i \mid Z_i = 0, W_i = w]}. \tag{9.8}$$

Unlike the scalar unconditional LATE, $CLATE(w)$ is a function. Flexible methods such as `grf::instrumental_forest()` can estimate heterogeneous conditional effects under suitable conditions.

9.3 Instrumental-variable estimation

The two moment equations associated with Equation (9.7) are

$$\mathbb{E}[Y_i - \beta_0 - \beta_1 D_i] = 0, \tag{9.9}$$
$$\mathbb{E}[Z_i(Y_i - \beta_0 - \beta_1 D_i)] = 0. \tag{9.10}$$

Eliminating the intercept gives

$$\beta_1 = \frac{\mathrm{Cov}(Y_i, Z_i)}{\mathrm{Cov}(D_i, Z_i)}. \tag{9.11}$$

For $p = \Pr(Z_i = 1)$,

$$\mathrm{Cov}(D_i, Z_i) = p(1-p)\{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]\},$$

so the denominator is nonzero exactly when $0 < p < 1$ and the instrument is relevant. Replacing moments by sample averages yields

$$\widehat{\beta}_{IV} = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum_{i=1}^{n}(D_i - \bar{D})(Z_i - \bar{Z})}. \tag{9.12}$$

With one binary instrument and one endogenous binary treatment, this is the sample analog of the Wald ratio in Equation (9.6).

9.3.1 Weak instruments

When the first-stage difference is close to zero, the ratio in Equation (9.12) is unstable. Weak instruments can produce substantial finite-sample bias and a poor normal approximation. The familiar "first-stage $F > 10$" rule is only a rough historical diagnostic, not a general guarantee. It depends on the number of instruments, the estimand, and the desired tolerance for bias or size distortion. Empirical work should report the first stage and use weak-instrument-robust diagnostics or confidence sets when weakness is a concern.

9.4 Where do instruments come from?

Instrumental variables transformed empirical economics, but a software command cannot make an instrument credible. Every application must defend both the exclusion restriction and relevance.

9.4.1 Randomized offers

RCT assignment is an especially transparent instrument when actual treatment has noncompliance. For example, Kline and Walters (2016) use randomized offers in the Head Start Impact Study to learn about early-childhood programs in the presence of close substitutes. The lottery identifies the effect for families whose program participation changes because of the offer.

9.4.2 Natural and quasi-experiments

Nature or policy can generate plausibly exogenous variation. Miguel, Satyanath, and Sergenti (2004), for example, use rainfall shocks as an instrument for economic growth when studying civil conflict in agriculturally dependent African countries. The exclusion argument is demanding: Sarsons (2015) shows why rainfall may be related to conflict through channels other than local income. This exchange illustrates an important habit—actively search for alternative pathways from Z to Y.

Policy expansions in eligibility for maternal care provide another possible instrument for service use. Staggered implementation may create relevance, but selective migration, concurrent programs, or direct income effects can threaten exclusion.

9.4.3 Variables excluded from the outcome equation

When no experiment is available, researchers sometimes use a variable believed to shift treatment but not the outcome directly. Classic returns-to-schooling examples include quarter of birth (Angrist and Krueger, 1991) and proximity to college (Card, 1995). Both require contestable institutional arguments: season of birth and place of residence may be related to family background, and their first-stage effects may be weak.

Other common constructions include reference-group variables and "Hausman-type" instruments, such as prices of the same product in other markets. Aggregation or geographic distance does not automatically guarantee exogeneity; correlated demand shocks can invalidate the instrument.

9.5 R implementation

The `ivreg` package uses the formula `outcome ~ treatment | instrument`. The following simulation contains never-takers, compliers, and always-takers but no defiers. The treatment effect is heterogeneous, so OLS and IV need not target the same parameter.

```
library(ivreg)

set.seed(5280)
n <- 4000
Z <- rbinom(n, 1, 0.5)
type <- sample(c("never", "complier", "always"), n,
               replace = TRUE, prob = c(0.25, 0.55, 0.20))

D0 <- as.integer(type == "always")
D1 <- as.integer(type %in% c("complier", "always"))
D <- ifelse(Z == 1, D1, D0)

U <- rnorm(n)
tau <- 1 + 0.7 * (type == "complier") + 0.4 * U
Y0 <- 0.5 + U + rnorm(n)
Y <- Y0 + D * tau

outcome_itt <- mean(Y[Z == 1]) - mean(Y[Z == 0])
first_stage <- mean(D[Z == 1]) - mean(D[Z == 0])
wald <- outcome_itt / first_stage
true_late <- mean(tau[type == "complier"])

c(outcome_itt = outcome_itt,
  first_stage = first_stage,
  wald = wald,
  true_late = true_late)

fit_iv <- ivreg(Y ~ D | Z)
summary(fit_iv, diagnostics = TRUE)
summary(lm(Y ~ D))
```

The companion Quarto Live page runs this example entirely in the browser. In real applications, diagnostics supplement rather than replace the design-based arguments for random assignment, exclusion, monotonicity, and relevance.

Summary

- Random assignment identifies the ITT effect of an offer even when treatment receipt is endogenous.
- With exclusion, relevance, and monotonicity, the outcome ITT divided by the first-stage ITT identifies LATE for compliers.
- IV is a ratio estimator based on the moment condition $\mathbb{E}[Z_i \varepsilon_i] = 0$; OLS fails when $\text{Cov}(D_i, \varepsilon_i) \neq 0$.

- Weak first stages make conventional normal approximations unreliable.
- A credible instrument requires an institutional argument, not merely a significant first stage.

Chapter 10

Neural Networks

Motivation. Obtain a glimpse of how deep learning can enter modern econometrics.

Neural networks. Build a flexible global approximation from simple layers.

Conditional expectations. Connect squared-error training to nuisance functions used in causal inference.

Application. Introduce DeepIV and large-scale heterogeneous position effects in sponsored search.

10.1 Neural networks as function approximators

Neural networks are the basic building blocks of deep learning. They are used for many tasks, but our focus is function approximation. CATE and CLATE are functions of covariates, while ATE, ATT, and LATE estimators often depend on conditional means or probabilities. Earlier chapters used regression trees and forests to estimate such functions. Neural networks provide a global, smooth alternative.

Consider $x = (x_1, x_2, x_3)'$ and an unknown real-valued function $h(x)$. A one-hidden-layer network constructs a parameterized approximation $h_\theta(x)$, as illustrated in Figure 10.1.

Each circle is a *unit* or *neuron*. Add a constant input $x_0 = 1$. For hidden unit $j \in \{1, 2, 3\}$, define

$$a_j^{(2)} = f^{(1)}\left(\theta_{0j}^{(1)} + \sum_{k=1}^{3} \theta_{kj}^{(1)} x_k\right), \tag{10.1}$$

where the weighted sum is the input to the unit and $f^{(1)}$ is its activation function. The output is

$$h_\theta(x) = f^{(2)}\left(\theta_0^{(2)} + \sum_{j=1}^{3} \theta_j^{(2)} a_j^{(2)}\right). \tag{10.2}$$

The network in Figure 10.1 has $(3+1) \times 3 + (3+1) \times 1 = 16$ weights and intercepts.

A familiar activation is the logistic, or sigmoid, function

$$\sigma(z) = \frac{1}{1 + e^{-z}}. \tag{10.3}$$

Other common choices include the rectified linear unit $\max\{0, z\}$ and hyperbolic tangent. For a continuous outcome, the output activation is often the identity; for a binary probability, a sigmoid output is natural.

Example 10.1 (Approximating a binary maximum)**.** Let $x_1, x_2 \in \{0, 1\}$ and $h(x) = \max\{x_1, x_2\}$. The single sigmoid unit

$$h_\theta(x) = \sigma(-20 + 40x_1 + 40x_2)$$

is already an excellent approximation:

$$h_\theta(0,0) = \sigma(-20) \approx 0,$$
$$h_\theta(1,0) = h_\theta(0,1) = \sigma(20) \approx 1,$$

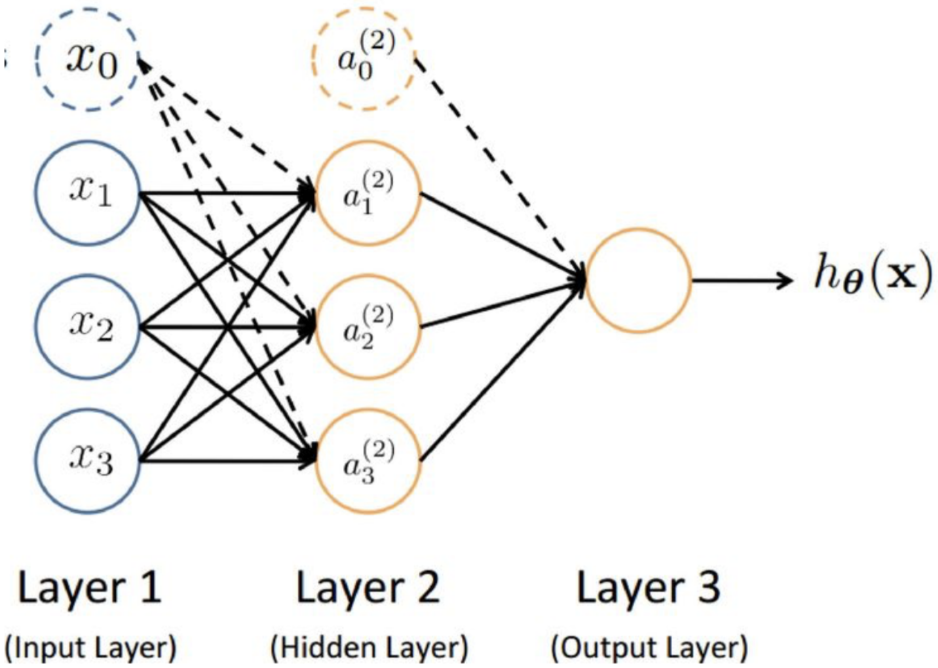

Figure 10.1: A one-hidden-layer feedforward neural network. The supplied course illustration is based on Andrew Ng's teaching materials.

$$h_\theta(1,1) = \sigma(60) \approx 1.$$

The original notes used inconsistent coefficient subscripts; the expression above uses the intercept and the two input coefficients explicitly.

Networks can be generalized in several directions:

- a hidden layer can contain many more units than the input dimension;
- several hidden layers can be stacked to form a deep network;
- the output can be a vector, as in multiclass prediction; and
- inputs can include images or text after they are numerically encoded.

Universal-approximation results show that, under suitable activation and regularity conditions, a one-hidden-layer network with sufficiently many units can approximate a continuous function on a compact set arbitrarily well. This is an approximation result, not a promise that finite data and an optimization algorithm will automatically find the best network.

For a visual experiment with architecture, activation functions, and training, see `https://playground.tensorflow.org`.

10.2 Conditional expectations as best predictors

The key link with econometrics is a projection identity. Suppose $\mathbb{E}[Y^2] < \infty$. Among all square-integrable measurable functions of W, the conditional expectation minimizes mean squared error:

$$m(W) = \mathbb{E}[Y \mid W] \in \arg\min_f \mathbb{E}[(Y - f(W))^2]. \tag{10.4}$$

Proof. Add and subtract $m(W)$ and expand:

$$\mathbb{E}[(Y - f(W))^2] = \mathbb{E}[(Y - m(W))^2] + \mathbb{E}[(m(W) - f(W))^2]$$

$$+2\mathbb{E}[(Y - m(W))(m(W) - f(W))].$$

The cross term is zero because iterated expectations give

$$\begin{aligned}
&\mathbb{E}[(Y - m(W))(m(W) - f(W))] \\
&\quad = \mathbb{E}[(m(W) - f(W))\mathbb{E}[Y - m(W) \mid W]] = 0.
\end{aligned}$$

The first term does not depend on f, and the second is nonnegative. It is minimized by $f(W) = m(W)$ almost surely. □

For treatment status $D \in \{0, 1\}$, define

$$m_d(w) = \mathbb{E}[Y \mid D = d, W = w]. \tag{10.5}$$

The corresponding population optimization problem is

$$m_d(\cdot) \in \arg\min_h \mathbb{E}[(Y - h(W))^2 \mid D = d]. \tag{10.6}$$

The argument of the minimum is a *function*; it is not the numerical conditional mean evaluated at one unspecified W.

10.3 From population loss to a trained network

Approximate $m_d(w)$ by $h_{d;\theta}(w)$. In the subsample with $D_i = d$, empirical risk minimization chooses

$$\widehat{\theta}_d \in \arg\min_\theta \frac{1}{n_d} \sum_{i:D_i=d} \big(Y_i - h_{d;\theta}(W_i)\big)^2, \qquad n_d = \sum_i \mathbf{1}\{D_i = d\}, \tag{10.7}$$

and the estimated conditional mean is $\widehat{m}_d(w) = h_{d;\widehat{\theta}_d}(w)$. A propensity score can be trained similarly with a binary cross-entropy loss and sigmoid output.

In practice, the numerical optimizer updates weights by backpropagation and a gradient-based method. Architecture, regularization, learning rate, number of epochs, and early stopping are tuning choices. Predictive performance should be assessed on validation data rather than on the same observations used to fit a highly flexible model.

10.3.1 Advantages

- Networks can model nonlinearities and interactions in high-dimensional W without specifying each one by hand.
- Categorical, text, and image inputs can be handled after appropriate numerical encoding.
- Matrix operations can be accelerated substantially on GPUs for large problems.

10.3.2 Limitations and inference

- Training is nonconvex and can be sensitive to scaling, architecture, initialization, and regularization.
- A highly accurate predictor need not extrapolate well or identify a causal parameter when the research design is invalid.
- Standard errors for every point of a flexible fitted CATE surface are not obtained by treating network weights like coefficients in a small linear regression.

Neural networks can nevertheless serve as nuisance estimators inside cross-fitted doubly robust or double-machine-learning procedures. If the nuisance estimates satisfy suitable rate and regularity conditions, the final low-dimensional ATE estimator can have valid asymptotic inference even though the individual network weights have no simple inferential interpretation.

10.4 Application: DeepIV

Hartford, Lewis, Leyton-Brown, and Taddy (2017) study sponsored-search position effects. An advertiser's position on a Bing search page is endogenous: latent user intent affects both placement and the probability of a click. Their DeepIV approach uses flexible neural networks in a nonparametric instrumental- variables procedure.

The instruments include indicators for randomized experiments in which advertiser–query pairs were assigned to different scoring algorithms. These experiments generate variation in advertisement position. Covariates include high-dimensional information such as the search query, and the sample exceeds 20 million observations. The empirical analysis finds, among other results, that a favorable position is especially valuable to small websites for on-brand queries.

The original implementation is available at `https://github.com/jhartford/DeepIV`. It is historically important, but software ecosystems evolve; contemporary replication should pin package versions and test the environment before class.

10.5 A small R illustration

The following code uses the lightweight `nnet` package to approximate a nonlinear conditional mean. It is deliberately small enough to run through Quarto Live in a browser.

```r
library(nnet)

set.seed(5280)
n <- 700
W1 <- runif(n, -1, 1)
W2 <- runif(n, -1, 1)
m <- sin(pi * W1) + W2^2
Y <- m + rnorm(n, sd = 0.25)

train <- sample(seq_len(n), 500)
dat <- data.frame(Y, W1, W2)

fit_nn <- nnet(Y ~ W1 + W2, data = dat[train, ],
               size = 8, linout = TRUE,
               decay = 0.01, maxit = 500, trace = FALSE)
fit_lm <- lm(Y ~ W1 + W2, data = dat[train, ])

test <- setdiff(seq_len(n), train)
pred_nn <- predict(fit_nn, dat[test, ])
pred_lm <- predict(fit_lm, dat[test, ])

c(neural_net = mean((Y[test] - pred_nn)^2),
  linear_model = mean((Y[test] - pred_lm)^2))
```

This comparison is about prediction, not causal identification. It shows only that a nonlinear architecture can approximate a nonlinear regression function more flexibly than a misspecified additive linear model.

Summary

- A neural network composes weighted sums and activation functions to form a flexible global approximation.
- The conditional expectation is the population squared-error minimizer, which connects prediction to causal nuisance functions.
- Training replaces the population loss by an empirical loss and requires tuning, regularization, and honest validation.
- Neural networks do not create causal identification, but they can be useful nuisance estimators inside valid causal designs and cross-fitted doubly robust procedures.

Chapter 11

Difference-in-Differences and Event Studies

Motivation. Policies are often adopted at different times in different places. Can these timing differences reveal causal effects?

Canonical DID. Identify an ATT with two groups, two periods, and parallel trends.

Staggered adoption. Define group–time effects when cohorts adopt in different periods.

Modern event studies. Avoid already-treated controls, estimate transparent effects first, and aggregate second.

Credibility and inference. Understand what pre-trends, confidence bands, and clustered standard errors can and cannot establish.

11.1 Motivation

Many important treatments in economics are policies rather than randomized experiments. A housing subsidy, minimum-wage increase, or new technology may be introduced in one place before another. A simple comparison between places with and without the policy is unlikely to be causal: adopters may have had different outcome levels before adoption. A before–after comparison within adopters is also insufficient: inflation, interest rates, or a common boom may have changed outcomes even without the policy.

Difference-in-differences (DID) combines both comparisons:

1. calculate the change over time for the treated group;
2. calculate the change for an untreated comparison group; and
3. subtract the second change from the first.

Permanent level differences disappear in the first difference. Common time changes disappear in the second. The remaining difference is causal under a parallel-trends assumption.

Example 11.1 (A two-by-two DID)**.** Suppose average monthly rent is

Group	Before	After	Change
Cities adopting a subsidy	120	134	14
Comparison cities	100	106	6

The DID is

$$(134 - 120) - (106 - 100) = 14 - 6 = 8.$$

Under parallel trends, 6 estimates how much rent would have risen in the adopting cities without the subsidy. The remaining 8 is the estimated effect on those cities.

11.2 Two groups and two periods

Let $t \in \{0, 1\}$ and define

- $G_i = 1$ if unit i belongs to the group treated beginning in period 1;
- $G_i = 0$ if it belongs to the never-treated control group; and
- $D_{it} = G_i \mathbf{1}\{t = 1\}$ as realized treatment status.

Let $Y_{it}(1)$ denote the potential outcome under the regime "treated beginning in period 1" and let $Y_{it}(0)$ denote the outcome under "never treated." The observed regime is selected by group:

$$Y_{it} = G_i Y_{it}(1) + (1 - G_i) Y_{it}(0). \tag{11.1}$$

The target is the post-treatment average effect on the treated,

$$ATT = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1]. \tag{11.2}$$

DID most directly learns about the group that actually adopts. The effect for a never-adopting group may differ and is not identified by the same comparison.

11.2.1 No anticipation

Before the policy begins, the future treatment regime should not yet affect the outcome:

$$Y_{i0}(1) = Y_{i0}(0). \tag{11.3}$$

This can fail if firms hire in advance of an announced subsidy or households move before a program formally starts. The last nominally pre-treatment period may then already be treated in an economic sense.

11.2.2 Parallel trends

The key identifying assumption is

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]. \tag{11.4}$$

This assumption concerns *changes*, not levels. Treated and control groups may have different average outcomes before treatment, and selection on permanent group differences is allowed. The assumption concerns the untreated potential outcome; after treatment, the treated group's counterfactual path is unobserved and cannot be checked directly.

11.2.3 Identification

Under no anticipation and parallel trends,

$$\begin{aligned} ATT &= \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] \\ &= \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0]. \end{aligned} \tag{11.5}$$

Every term in the final line is observed. The sample estimator is

$$\widehat{ATT}_{DID} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}), \tag{11.6}$$

where $\bar{Y}_{gt}$ is the sample mean for group g in period t. The order of differencing is immaterial:

$$(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) = (\bar{Y}_{11} - \bar{Y}_{01}) - (\bar{Y}_{10} - \bar{Y}_{00}).$$

11.2.4 Regression representation

With two groups and two periods, Equation (11.6) is the OLS coefficient $\widehat{\tau}$ in

$$Y_{it} = \beta_0 + \beta_1 G_i + \beta_2 Post_t + \tau(G_i \times Post_t) + \varepsilon_{it}, \tag{11.7}$$

where $Post_t = \mathbf{1}\{t = 1\}$. With panel data, the same coefficient comes from

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}, \tag{11.8}$$

where α_i and λ_t are unit and time fixed effects. In this canonical setting, two-way fixed effects (TWFE) is simply another way to calculate DID.

The same units are followed over time in panel data, whereas repeated cross sections sample different units in each period. Both can identify ATT under appropriate sampling and parallel-trends assumptions. Panel data are often more efficient because differencing removes permanent unit-specific noise.

11.3 Staggered treatment adoption

Now let $t = 1, \dots, T$. Define $G_i = g$ if unit i is first treated in period g, and $G_i = \infty$ if it is never treated. In R code, never-treated units are usually recorded as $G_i = 0$. With absorbing treatment,

$$D_{it} = \mathbf{1}\{t \geq G_i\} \quad \text{for eventually treated units,} \qquad D_{it} = 0 \quad \text{if } G_i = \infty. \tag{11.9}$$

Thus $D_{it} = 1$ implies $D_{i,t+1} = 1$. A treatment that repeatedly switches on and off requires a different design or additional assumptions.

Let $Y_{it}(g)$ be the outcome if unit i is first treated in period g, and let $Y_{it}(0)$ be the never-treated outcome. No anticipation requires

$$Y_{it}(g) = Y_{it}(0), \qquad t < g. \tag{11.10}$$

The natural causal building block is the group–time effect

$$ATT(g,t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g], \qquad t \geq g. \tag{11.11}$$

$ATT(4,4)$ is the adoption-period effect for cohort 4; $ATT(4,6)$ is the effect two periods later for the same cohort; and $ATT(6,6)$ concerns another cohort in the same calendar period. Policy effects can vary across all these cells.

11.3.1 Valid control groups

For $ATT(g,t)$, controls must be untreated in both the baseline $g-1$ and the comparison period t. Two choices are common:

Never-treated controls. Compare cohort g with units that never adopt.

Not-yet-treated controls. Compare cohort g with units treated after t, together with never-treated units.

The second choice uses more observations but assumes later adopters provide a valid counterfactual trend for earlier adopters. This is an identification choice, not merely a software option.

With never-treated controls, assume

$$\mathbb{E}[Y_{it}(0) - Y_{i,g-1}(0) \mid G_i = g] = \mathbb{E}[Y_{it}(0) - Y_{i,g-1}(0) \mid G_i = \infty]. \tag{11.12}$$

Then

$$ATT(g,t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = \infty]. \tag{11.13}$$

For not-yet-treated controls, replace $G_i = \infty$ by $G_i > t$ in both displays, using the convention that $\infty > t$. Each $ATT(g,t)$ is then a clean two-group, two-period DID. Already-treated observations are not used as untreated controls.

11.3.2 Conditional parallel trends and overlap

Let W_i contain pre-treatment covariates. A conditional assumption for not-yet-treated controls is

$$\begin{aligned} &\mathbb{E}[Y_{it}(0) - Y_{i,g-1}(0) \mid G_i = g, W_i] \\ &\quad = \mathbb{E}[Y_{it}(0) - Y_{i,g-1}(0) \mid G_i > t, W_i]. \end{aligned} \tag{11.14}$$

Overlap requires controls at every covariate value represented in cohort g:

$$\Pr(G_i > t \mid W_i = w) > 0 \quad \text{for } w \in \operatorname{supp}(W_i \mid G_i = g). \tag{11.15}$$

Under these conditions,

$$\begin{aligned} ATT(g,t) = \mathbb{E}\Big[&\mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g, W_i] \\ &- \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i > t, W_i] \;\Big|\; G_i = g\Big]. \end{aligned} \tag{11.16}$$

Outcome regression estimates conditional control trends; propensity weighting reweights controls toward cohort g; and a doubly robust DID combines both. Under the relevant regularity conditions, a doubly robust estimator remains consistent if either the outcome-trend model or the generalized propensity model is correct. Covariates should be measured before treatment, not caused by it.

11.4 Why conventional TWFE can fail

With many periods, researchers often estimate

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}. \tag{11.17}$$

Under staggered adoption and heterogeneous effects, β need not be an interpretable average treatment effect.

Imagine an early-treated group E, a late-treated group L, and a never-treated group N. When E is newly treated, L and N are untreated. When L is newly treated, E is already treated. Conventional TWFE can use all of the following:

1. newly treated versus never treated;
2. newly treated versus not yet treated; and
3. newly treated versus already treated.

The third comparison subtracts a change containing E's treatment-effect dynamics. It is not L's untreated counterfactual trend.

The Goodman–Bacon decomposition gives nonnegative weights to component two-by-two DID comparisons, but some components use already-treated controls. When the TWFE coefficient is instead written in terms of cohort–time treatment effects, implicit weights can be negative. In extreme cases, every causal effect can be positive while the static TWFE coefficient is negative.

11.4.1 Conventional event-study regressions

A familiar dynamic regression is

$$Y_{it} = \alpha_i + \lambda_t + \sum_{e \neq -1} \mu_e \mathbf{1}\{t - G_i = e\} + u_{it}, \tag{11.18}$$

where $e = t - G_i$ is event time and $e = -1$ is omitted. With staggered timing and heterogeneous effects, μ_e can be contaminated by effects at other event times. Even a lead can be nonzero because of post-treatment heterogeneity rather than a genuine pre-trend. Adding leads and lags to TWFE does not by itself repair the control-group problem.

Fixed effects remain useful. TWFE equals DID in the canonical two-by-two design; a single treated cohort with a valid never-treated group avoids already-treated controls; and a homogeneous common-effect model can justify a static coefficient. The lesson is to define the causal parameter and valid controls before running the regression.

11.5 Modern DID and event studies

Important

Estimate transparent cohort–time causal effects using untreated controls, and only then aggregate them into the summary required by the empirical question.

The Callaway–Sant'Anna approach estimates the collection of $ATT(g, t)$ values using clean comparisons. Depending on the design and covariates, each cell can be estimated by differences in outcome changes, outcome regression, inverse probability weighting, or doubly robust DID.

11.5.1 Aggregating group–time effects

One overall summary is

$$\theta^{overall} = \sum_{g} \sum_{t \geq g} \omega_{g,t} ATT(g, t), \qquad \omega_{g,t} \geq 0, \quad \sum_{g,t} \omega_{g,t} = 1. \tag{11.19}$$

Other summaries average within adoption cohort or within calendar period. They answer different questions. Software cannot decide which is economically relevant.

In the `did` package, `type = "simple"` weights post-treatment cells in proportion to cohort size, so early cohorts receive more total weight because they contribute more cells. `type = "group"` first averages post-treatment effects within cohort. `type = "dynamic"` averages contributing cohorts at each event time, using cohort-size weights.

For event time $e = t - g$, define

$$\theta(e) = \sum_{g: g+e \leq T} \Pr(G_i = g \mid G_i + e \leq T) ATT(g, g + e). \tag{11.20}$$

For $e \geq 0$, this is the average effect after e periods of exposure among cohorts observed at that horizon. At long horizons only early adopters remain, so a changing curve can reflect both genuine dynamics and changing cohort composition. A balanced event window holds the contributing cohorts fixed, improving comparability at the cost of discarding data.

Other modern approaches implement a similar principle in different ways:

Sun–Abraham. Estimate cohort-by-event-time interactions and aggregate them without contamination from other event times.

Borusyak–Jaravel–Spiess. Fit an untreated-outcome model using untreated observations, impute untreated counterfactuals for treated cells, and average the resulting effects.

Group–time DID. Construct clean two-by-two comparisons for each (g, t) and then aggregate.

These methods can target different parameters and maintain different outcome, comparison-group, or covariate models. They are not interchangeable commands.

11.6 Credibility and inference

Modern event studies often report placebo or pseudo-ATT estimates before treatment. Large, systematic pre-treatment differences weigh against the proposed parallel-trends design. Estimates near zero do not prove it:

- pre-treatment estimates may be noisy and have low power;
- paths may be parallel before treatment but diverge afterward;
- choosing a specification because its pre-trend test is insignificant changes the statistical behavior of the final estimates; and
- the assumption concerns an unobserved post-treatment counterfactual.

Pre-trends are a diagnostic, not a certificate of validity.

An event study reports many coefficients. Pointwise 95% intervals cover each coefficient separately, not the complete curve jointly. For statements such as "all pre-treatment effects are zero," simultaneous confidence bands are preferable. Modern DID packages can obtain uniform bands by multiplier bootstrap.

Outcomes from the same unit are dependent over time. When treatment is assigned at a higher level, observations may also be dependent within that assignment cluster. With state-level policies, cluster at the state level rather than the individual level. With few clusters, ordinary cluster-robust approximations may be unreliable. The clustering level follows treatment assignment and the dependence structure, not the number of rows.

11.6.1 A design checklist

Before estimating a staggered DID, ask:

1. What is the unit, and when is it first treated?
2. Is treatment absorbing, or can it turn off?
3. Could units anticipate treatment?
4. Which observations are valid controls for each cohort and period?
5. Is parallel trends more plausible conditionally on pre-treatment covariates?
6. Is the target group–time, dynamic, calendar-time, or overall ATT?
7. Does cohort composition change across the event window?
8. At what level should inference be clustered?
9. Do other policies or shocks occur at the same time?
10. Are conclusions robust to outcome scale, sample window, control group, and anticipation window?

11.7 R implementation

We simulate a balanced panel in which cohorts are first treated in periods 4, 6, or 8, while some units are never treated. Treatment effects grow with exposure and differ across cohorts—exactly the setting in which conventional TWFE can be hard to interpret.

11.7.1 Generate the panel

```
set.seed(5280)
N <- 600
TT <- 10

units <- data.frame(
  id = 1:N,
  G = sample(c(0, 4, 6, 8), N, replace = TRUE,
             prob = c(0.30, 0.25, 0.25, 0.20)),
  alpha = rnorm(N),
  W = rnorm(N)
)

data <- merge(expand.grid(id = 1:N, t = 1:TT), units,
              by = "id", sort = TRUE)
data$D <- as.integer(data$G > 0 & data$t >= data$G)
data$event_time <- ifelse(data$G > 0, data$t - data$G, NA)

data$cohort_scale <- ifelse(data$G == 4, 1.50,
                     ifelse(data$G == 6, 1.00,
                     ifelse(data$G == 8, 0.60, 0)))
data$tau <- ifelse(data$D == 1,
                   data$cohort_scale * (1 + 0.5 * data$event_time), 0)

data$lambda <- 0.25 * data$t + 0.03 * data$t^2
data$Y0 <- data$alpha + data$lambda + 0.5 * data$W +
           rnorm(nrow(data))
data$Y <- data$Y0 + data$tau
```

Untreated outcomes satisfy parallel trends by construction. The treatment effect is positive but heterogeneous across cohorts and exposure lengths.

11.7.2 Calculate one group–time DID by hand

```
simple_att_gt <- function(data, g, t) {
  stopifnot(t >= g)
  base <- g - 1

  now <- data[data$t == t, c("id", "G", "Y")]
  names(now)[names(now) == "Y"] <- "Y_now"
  before <- data[data$t == base, c("id", "Y")]
  names(before)[names(before) == "Y"] <- "Y_before"

  change <- merge(now, before, by = "id")
  change$dY <- change$Y_now - change$Y_before
  treated <- change$G == g
  controls <- change$G == 0 | change$G > t
```

```r

  mean(change$dY[treated]) - mean(change$dY[controls])
}

simple_att_gt(data, g = 4, t = 6)
mean(data$tau[data$G == 4 & data$t == 6])
```

The two numbers should be close. The second is available only because the DGP is known; in real data it is the unobserved causal parameter.

11.7.3 Callaway–Sant'Anna group–time effects

```r
library(did)

att <- att_gt(
  yname = "Y", tname = "t", idname = "id", gname = "G",
  xformla = ~ 1, data = data, panel = TRUE,
  control_group = "notyettreated", est_method = "dr",
  base_period = "universal", clustervars = "id",
  bstrap = TRUE, cband = TRUE,
  biters = 199                 # short browser demonstration only
)

summary(att)
ggdid(att)
```

With `xformla = ~ 1`, the built-in outcome-regression, IPW, and doubly robust implementations have the same unconditional two-by-two DID point estimates. Replace it by, for example, `~ W` for a conditional design.

The universal base period uses $g - 1$ throughout and normalizes event time -1 to zero. The varying-base default leaves post-treatment effects unchanged but uses adjacent-period comparisons for pre-treatment pseudo-ATTs. The multiplier bootstrap creates simultaneous bands. We use only 199 repetitions to keep a live classroom run short; use at least 999, and often more, for final empirical inference. If treatment is assigned above the unit level, specify the relevant assignment cluster. For panel data, `did` permits at most two cluster variables and requires one to be the unit identifier.

11.7.4 Aggregate the effects

```r
overall <- aggte(att, type = "simple")
by_group <- aggte(att, type = "group")
dynamic <- aggte(att, type = "dynamic", min_e = -3, max_e = 4)

summary(overall)
summary(by_group)
summary(dynamic)
ggdid(dynamic)

dynamic_balanced <- aggte(
  att, type = "dynamic",
  min_e = -3, max_e = 3,
  balance_e = 3
)
```

```
summary(dynamic_balanced)
ggdid(dynamic_balanced)
```

The simple, group, and dynamic outputs answer different questions and need not be numerically equal.

11.7.5 Conventional and interaction-weighted event studies

```
library(fixest)

data$ever_treated <- as.integer(data$G > 0)
data$event_twfe <- ifelse(data$G > 0, data$t - data$G, 0)

twfe_event <- feols(
  Y ~ i(event_twfe, ever_treated, ref = -1) | id + t,
  data = data, cluster = ~ id
)

# A cohort outside the observed period range is treated as never treated.
data$G_sunab <- ifelse(data$G == 0, 1000, data$G)
sa_event <- feols(
  Y ~ sunab(G_sunab, t, ref.p = -1) | id + t,
  data = data, cluster = ~ id
)

iplot(list("Conventional TWFE" = twfe_event,
           "Sun--Abraham" = sa_event),
      ref.line = -1, main = "Event-study estimates")
```

The fixest::iplot() intervals here are pointwise. They are not the multiplier-bootstrap simultaneous bands shown by ggdid().

The companion page pins WebR 0.5.9 (R 4.5), for which the WebAssembly builds of did, its fastglm dependency, and fixest are mutually compatible. This is a runtime compatibility pin, not an econometric choice.

Summary

- DID compares outcome changes, not outcome levels.
- No anticipation and parallel trends for untreated potential outcomes identify the canonical two-by-two ATT.
- Under staggered adoption, $ATT(g,t)$ is the transparent causal building block; never-treated or not-yet-treated units can be controls.
- Conventional static and dynamic TWFE can be contaminated by already-treated comparisons when effects are heterogeneous.
- Modern procedures estimate valid cohort–time comparisons first and aggregate them second.
- Pre-trends, aggregation, cohort composition, anticipation, covariates, and clustering are substantive design choices.

Further reading

- Callaway and Sant'Anna (2021), "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, `https://doi.org/10.1016/j.jeconom.2020.12.001`.
- Sun and Abraham (2021), "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, `https://doi.org/10.1016/j.jeconom.2020.09.006`.
- Goodman-Bacon (2021), "Difference-in-Differences with Variation in Treatment Timing," *Journal of Econometrics*, `https://doi.org/10.1016/j.jeconom.2021.03.014`.
- Borusyak, Jaravel, and Spiess (2024), "Revisiting Event-Study Designs: Robust and Efficient Estimation," *Review of Economic Studies*, `https://doi.org/10.1093/restud/rdae007`.
- Roth, Sant'Anna, Bilinski, and Poe (2023), "What's Trending in Difference-in-Differences?," *Journal of Econometrics*, `https://doi.org/10.1016/j.jeconom.2023.03.008`.
- Sant'Anna and Zhao (2020), "Doubly Robust Difference-in-Differences Estimators," *Journal of Econometrics*, `https://doi.org/10.1016/j.jeconom.2020.06.003`.
- Rambachan and Roth (2023), "A More Credible Approach to Parallel Trends," *Review of Economic Studies*, `https://doi.org/10.1093/restud/rdad018`.

Appendix A

Additional Matrix Algebra

This appendix collects material that is useful for reference but is not needed before the main econometrics chapters.

A.1 Addition and scalar multiplication

Matrices can be added only when they have the same dimensions, and addition is entry by entry. If A and B are $m \times k$, then

$$(A + B)_{ij} = a_{ij} + b_{ij}.$$

For a scalar c, scalar multiplication is also entry by entry: $(cA)_{ij} = ca_{ij}$. Consequently,

$$A + B = B + A, \qquad c(A + B) = cA + cB.$$

A.2 Several views of multiplication

Let A be $n \times k$, B be $n \times p$, and $C = A'B$. In addition to the entry formula in Chapter 2, we can write

$$C_{ab} = A'_{\cdot a} B_{\cdot b} \quad \text{and} \quad A'B = \sum_{i=1}^{n} A_i B_i',$$

where $A_{\cdot a}$ denotes column a of A, while A_i and B_i are the column-vector versions of row i. Each term $A_i B_i'$ is $k \times p$.

The following code verifies the two representations. It is pedagogically useful, although in practice one should use R's optimized matrix product.

```
A <- matrix(c(1, 2, 3, 4, 5, 6), nrow = 3, ncol = 2)
B <- matrix(c(2, 3, 4, 5, 6, 7), nrow = 3, ncol = 2)

# Direct product
C <- t(A) %*% B

# Entry-by-entry representation
C_entry <- matrix(0, ncol(A), ncol(B))
for (a in seq_len(ncol(A))) {
  for (b in seq_len(ncol(B))) {
    C_entry[a, b] <- sum(A[, a] * B[, b])
  }
}

# Sum of row outer products
C_outer <- matrix(0, ncol(A), ncol(B))
for (i in seq_len(nrow(A))) {
  ai <- matrix(A[i, ], ncol = 1)
  bi <- matrix(B[i, ], nrow = 1)
  C_outer <- C_outer + ai %*% bi
}
```

```

C
C_entry
C_outer
```

A.3 Rank and inverse formulas

Some useful rank facts are

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$$

and

$$\text{rank}(A'A) = \text{rank}(AA') = \text{rank}(A).$$

The second equality does *not* imply that both $A'A$ and AA' are invertible when A is rectangular. If A is $m \times k$ with $m > k$ and full column rank, then $A'A$ is invertible but AA' has rank $k < m$ and is singular. The roles reverse when $m < k$ and A has full row rank.

For an invertible diagonal matrix $D = \text{diag}(d_1, \ldots, d_k)$,

$$D^{-1} = \text{diag}(d_1^{-1}, \ldots, d_k^{-1}).$$

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is $ad - bc$, and, when $ad - bc \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

A.4 Eigenvalues and spectral decomposition

An eigenvalue–eigenvector pair (λ, v) of a square matrix A satisfies $Av = \lambda v$ with $v \neq 0$. A general real square matrix need not have a full set of real eigenvectors and need not be diagonalizable. For example, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not diagonalizable.

The clean result needed most often in econometrics applies to symmetric matrices.

Theorem A.1 (Spectral theorem)**.** *For every real symmetric $k \times k$ matrix A, there are an orthogonal matrix H and a real diagonal matrix Λ such that*

$$A = H\Lambda H', \qquad H'H = HH' = I_k.$$

The diagonal entries of Λ are the eigenvalues of A, and the columns of H are orthonormal eigenvectors.

The decomposition is not unique: eigenvectors can be reordered, their signs can be changed, and within an eigenspace associated with a repeated eigenvalue, any orthonormal basis may be used.

For a real symmetric matrix:

- A is p.s.d. if and only if every eigenvalue is nonnegative;
- A is p.d. if and only if every eigenvalue is positive;
- $\text{rank}(A)$ equals the number of nonzero eigenvalues;

- A is invertible if and only if zero is not an eigenvalue, and then $A^{-1} = H\Lambda^{-1}H'$.

The rank statement is not valid for every nonsymmetric matrix. A nilpotent matrix such as $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has rank one but only the eigenvalue zero.

For a rectangular matrix, the analogous tool is the singular-value decomposition $A = U\Sigma V'$. Its rank equals the number of nonzero singular values. This is why singular values, rather than ordinary eigenvalues, are the safe general-purpose rank diagnostic.

```r
library(Matrix)
A <- matrix(c(2, 1, 1, 2), 2, 2)
eigen(A)
rankMatrix(A)
solve(A)

# A nonsymmetric matrix that is not diagonalizable
J <- matrix(c(1, 0, 1, 1), 2, 2)
eigen(J)
```

A.5 Matrix derivatives

We use the convention that the gradient of a scalar-valued function with respect to a column vector is itself a column vector. For conformable fixed objects x, y, and X,

$$\nabla_\beta(x'\beta) = x, \qquad \nabla_\beta(\beta'\beta) = 2\beta, \qquad \frac{\partial(X\beta)}{\partial\beta'} = X.$$

The last object is a Jacobian, not a scalar gradient.

For the least-squares criterion

$$Q(\beta) = (Y - X\beta)'(Y - X\beta),$$

expansion or the chain rule gives

$$\begin{aligned} \nabla_\beta Q(\beta) &= -2X'(Y - X\beta) \\ &= 2X'(X\beta - Y). \end{aligned}$$

The minus sign is essential. Setting the gradient to zero yields the normal equations $X'X\beta = X'Y$, and if X has full column rank, $\widehat{\beta} = (X'X)^{-1}X'Y$.

A.6 Further reading

Useful extensions include determinants, traces, idempotent projection matrices, generalized inverses, and singular-value decomposition. A compact reference is Petersen and Pedersen's *Matrix Cookbook*.

Appendix B

Additional Probability and Asymptotic Tools

This appendix provides a more careful probability-space description, a few distributional examples, simulation code for the WLLN and CLT, and the delta method. None of it is required on a first reading of Chapter 3.

B.1 Probability spaces, events, and random variables

A probability space is a triple $(\mathcal{S}, \mathcal{F}, P)$:

- $\mathcal{S}$ is the *sample space* of elementary outcomes;
- $\mathcal{F}$ is a sigma-algebra of events, meaning it contains $\mathcal{S}$ and is closed under complements and countable unions;
- $P : \mathcal{F} \to [0, 1]$ satisfies $P(\mathcal{S}) = 1$ and countable additivity over pairwise disjoint events.

A real random variable is a measurable map $X : \mathcal{S} \to \mathbb{R}$. Its distribution is the probability measure induced on the real line:

$$P_X(B) = P\{s \in \mathcal{S} : X(s) \in B\}.$$

Thus, the sample space contains underlying outcomes, whereas the support of X contains possible numerical values. Treating those two sets as the same is a convenient shortcut in elementary examples, but it is not the general definition.

For example, an outcome can record tomorrow's temperature, rainfall, and wind. The random variable X may retain only the indicator that it rains. Then the sample space is rich, while the support of X is merely $\{0, 1\}$.

B.2 CDFs, densities, and continuous distributions

Every real random variable has a right-continuous c.d.f. $F_X(x) = P(X \le x)$. A variable is *continuous* (atomless) if $P(X = x) = 0$ for every x, equivalently if its c.d.f. is continuous. It is *absolutely continuous* if there exists an integrable density f_X such that

$$F_X(x) = \int_{-\infty}^{x} f_X(s)\, ds.$$

Every absolutely continuous distribution is continuous, but the converse is false. The Cantor distribution is the standard counterexample: it has a continuous c.d.f. but no density with respect to ordinary length (Lebesgue measure).

The geometric "probability equals length" picture is therefore not a general definition. If X is uniform on $[0, 1]$, then $P(X \in A)$ equals the length of A for suitable sets A. Under a nonuniform density, probability is instead weighted length, $P(X \in A) = \int_A f_X(x)\, dx$; for a discrete distribution, it is a sum of point masses.

The code below compares three normal densities. It uses base R so that the live page loads quickly.

```
x <- seq(-7, 7, length.out = 400)
plot(x, dnorm(x, 0, 1), type = "l", lwd = 2,
     xlab = "x", ylab = "density", ylim = c(0, 0.42))
lines(x, dnorm(x, 2, 1), lty = 2, lwd = 2, col = "steelblue")
lines(x, dnorm(x, 0, 2), lty = 3, lwd = 2, col = "firebrick")
legend("topright", c("N(0,1)", "N(2,1)", "N(0,4)"),
       lty = 1:3, lwd = 2,
       col = c("black", "steelblue", "firebrick"), bty = "n")
```

B.3 Conditional distributions with continuous variables

When $P(X = x) = 0$, elementary event conditioning cannot define $P(Y \leq y \mid X = x)$ by a ratio of probabilities. A regular conditional distribution supplies a mathematically valid version. When a joint density exists and $f_X(x) > 0$, it has the familiar formula

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

One sometimes motivates this expression using limits of shrinking intervals around x. Such a limit is an intuition, not a universal definition: its validity requires regularity conditions and it identifies the conditional law only for almost every x.

Bayes' rule for densities is

$$f_{X|Y}(x \mid y) = \frac{f_{Y|X}(y \mid x) f_X(x)}{f_Y(y)}, \qquad f_Y(y) > 0.$$

Marginalization gives

$$f_X(x) = \int f_{X,Y}(x,y)\, dy, \qquad F_X(x) = \int F_{X|Y}(x \mid y)\, dF_Y(y).$$

B.4 The bivariate normal example

A random vector $(X,Y)'$ is jointly normal if every linear combination $aX + bY$ is normally distributed. For a jointly normal pair, zero covariance is equivalent to independence. This equivalence is special: it is false for general distributions.

The following optional demonstration plots bivariate normal densities with positive, negative, and zero correlation. The live version uses `mvtnorm` and `plotly`; the two-dimensional contour option is usually quicker in class.

```
library(mvtnorm)
x <- seq(-4, 4, length.out = 80)
y <- seq(-4, 4, length.out = 80)
mu <- c(0, 0)
Sigma <- matrix(c(1, 0.5, 0.5, 1), 2, 2)
z <- outer(x, y, Vectorize(function(a, b) {
  dmvnorm(c(a, b), mean = mu, sigma = Sigma)
}))
contour(x, y, z, xlab = "X", ylab = "Y")
```

B.5 A compact WLLN and CLT simulation

The original classroom simulation repeated similar commands for ten sample sizes. The vectorized version below does the same job more transparently. For Bernoulli(0.5) observations, the WLLN predicts concentration of $\bar{X}_n$ around 0.5, and the CLT predicts

$$\frac{\sqrt{n}(\bar{X}_n - 0.5)}{0.5} \xrightarrow{d} N(0,1),$$

because the Bernoulli standard deviation is 0.5.

```r
set.seed(12)
B <- 1000
n_grid <- c(10, 25, 50, 100, 250)

sim_one_n <- function(n) {
  means <- replicate(B, mean(rbinom(n, 1, 0.5)))
  data.frame(
    n = n,
    mean = means,
    standardized = sqrt(n) * (means - 0.5) / 0.5
  )
}
sim <- do.call(rbind, lapply(n_grid, sim_one_n))

# WLLN: estimated probability of falling within 0.1 of the mean
aggregate(abs(mean - 0.5) < 0.1 ~ n, sim, mean)

# CLT: compare one standardized histogram with N(0,1)
z <- subset(sim, n == max(n_grid))$standardized
hist(z, probability = TRUE, breaks = 25,
     main = paste("n =", max(n_grid)), xlab = "standardized mean")
curve(dnorm(x), add = TRUE, lwd = 2, col = "firebrick")
```

B.6 The delta method

The delta method transfers an asymptotic distribution through a smooth transformation. It is a first-order Taylor expansion with probabilistic remainder control.

Theorem B.1 (Multivariate delta method). *Suppose $X_n \in \mathbb{R}^k$ satisfies*

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \Sigma),$$

and $g : \mathbb{R}^k \to \mathbb{R}^\ell$ is continuously differentiable near θ. Let $G(\theta)$ be the $\ell \times k$ Jacobian whose (r, j) entry is $\partial g_r(\theta)/\partial\theta_j$. Then

$$\sqrt{n}\{g(X_n) - g(\theta)\} \xrightarrow{d} N\big(0, G(\theta)\Sigma G(\theta)'\big).$$

For a scalar-valued g, $G(\theta)$ is a row vector and the limiting variance is $G(\theta)\Sigma G(\theta)'$. A statement such as "converges to $N(0, \cdot)$" must include the N; a covariance matrix alone is not a distribution. If the first derivative is zero, this first-order result is degenerate and a higher-order delta method may be needed.

Example B.1 (Log transformation)**.** If $\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, V)$ with $\theta > 0$, then $g(\theta) = \log \theta$ has derivative $1/\theta$, so

$$\sqrt{n}\{\log \widehat{\theta} - \log \theta\} \xrightarrow{d} N\left(0, \frac{V}{\theta^2}\right).$$

Runtime and Reproducibility

The source bundle includes a Quarto Live/WebR site and a two-core GitHub Codespaces configuration. Package declarations are kept chapter-specific to reduce browser startup time. See the project `README.md` for build and deployment instructions.